Forced Magnetic Reconnection in Tokamak Plasmas

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I. INTRODUCTION

Consider a tokamak plasma that is subject to an externally generated, non-rotating, non-axisymmetric, magnetic perturbation that is switched on suddenly. The perturbation inevitably resonates at various rational surfaces inside the plasma, and acts to drive magnetic reconnection at these surfaces. This process, which is known as *forced magnetic reconnection*, generates magnetic island chains within the plasma. These chains degrade the energy confinement properties of the plasma, and can trigger a major disruption.[1,2]

This paper investigates the response of an idealized cylindrical tokamak plasma to a non-rotating resonant magnetic perturbation that is switched on suddenly. The problem under discussion is an extension of one considered in a well-known paper by Hahm & Kulsrud,[3] and a host of subsequent papers.[4–11] The present study takes plasma flows, including diamagnetic flows, into account. Such flows are often omitted from the analysis, but can profoundly modify forced magnetic reconnection. The ion sound radius, and realistic levels of perpendicular energy and momentum transport, are also included in the analysis.

II. PRELIMINARY ANALYSIS

A. Plasma Equilibrium

Our tokamak equilibrium is approximated as a periodic cylinder of circular cross-section. Let us employ a conventional set of right-handed cylindrical coordinates, r, θ, z. The equilibrium magnetic flux-surfaces lie on surfaces of constant r. The system is assumed to be periodic in the z ('toroidal') direction, with periodicity length $2\pi\,R_0$, where R_0 is the simulated major radius of the plasma. It is helpful to define the simulated toroidal angle $\varphi = z/R_0$. Let a be the minor radius of the plasma. It is assumed that $a/R_0 \ll 1$. It is also assumed that the toroidal plasma beta parameter is $\mathcal{O}(a^2/R_0^2)$. The equilibrium magnetic field is written $\mathbf{B} = (0,\, B_\theta(r),\, B_\varphi)$, where $B_\theta(r)$ is the poloidal magnetic field-strength, and B_φ the (approximately) spatially uniform toroidal magnetic field-strength. The equilibrium current density takes the form $\mathbf{j} = (0,\, 0,\, j_\varphi(r))$. The *safety-factor* profile is defined as

$$q(r) = \frac{r\,B_\varphi}{R_0\,B_\theta}. \tag{1}$$

B. Plasma Response to Resonant Magnetic Perturbation

Suppose that the plasma equilibrium is subject to a relatively small amplitude (compared to the equilibrium magnetic field) helical magnetic perturbation with m periods in the poloidal direction, and n periods in the poloidal direction which is switched on suddenly at $t = 0$. The perturbation resonates with the plasma at the magnetic flux-surface at which $\mathbf{k}\cdot\mathbf{B} = 0$, where $\mathbf{k} = (0,\, m/r, -n/R_0)$ is the wavenumber of the perturbation. It follows that the safety-factor takes the value m/n at this so-called *rational* surface. As far as its response to the magnetic perturbation is concerned, a high Lundquist number tokamak plasma can be divided into two regions.[1–5,8,10–14] The so-called *inner region* lies in the immediate vicinity of the rational surface. The so-called *outer region* comprises everywhere in the plasma apart from the inner region, as well as the vacuum region surrounding the plasma. The response of the plasma to the magnetic perturbation in the outer region is governed by the equations of linear, marginally-stable, ideal-magnetohydrodynamics.[1] The minimal model that can realistically describe the response of the plasma in the inner region should contain plasma resistivity, E-cross-B flows, electron and ion diamagnetic flows, the ion sound radius, and anomalous cross-flux-surface transport of energy and momentum. It is important to include diamagnetic flows in the model because they are likely to be similar in magnitude to E-cross-B flows in next-generation tokamaks. It is important to include the ion sound radius in the model because the response of the ion fluid to the magnetic perturbation is radically different to that of the electron fluid on length-scales below the ion sound radius. Finally, it is important to include realistic levels of perpendicular energy and momentum transport in the model because these effects significantly change the response of the inner region to the magnetic perturbation.[13] The solutions in the inner and outer regions must be asymptotically matched to one another.[12]

Plasma flows in the inner region lead to the development of a localized *shielding current* that suppresses driven magnetic reconnection at the rational surface. However, the magnetic perturbation exerts a localized electromagnetic torque in the inner region that acts to modify the plasma flows in such a manner as to reduce the shielding current. As the amplitude of the perturbation gradually increases in time, a critical amplitude is eventually reached at which there is a sudden shift in the plasma flows, leading to the complete elimination of the shielding current, and allowing unhindered driven magnetic reconnection.[1,2,11,13,14]

III. OUTER REGION

A. Cylindrical Tearing Mode Equation

In the outer region, the perturbed magnetic flux associated with the plasma response can be written [1]

$$\delta\psi(r,\zeta,t) = \psi_1(r,t)\,\mathrm{e}^{\,\mathrm{i}\,\zeta}, \tag{2}$$

where $\zeta = m\,\theta - n\,\varphi$. The associated perturbed magnetic field is [1]

$$\delta B_r = \mathrm{i}\,\frac{m}{r}\,\delta\psi, \tag{3}$$

$$\delta B_\theta = -\frac{\partial\delta\psi}{\partial r}, \tag{4}$$

$$\delta B_\varphi \simeq 0. \tag{5}$$

The function $\psi_1(r,t)$ satisfies the *cylindrical tearing mode equation*,[1]

$$\frac{\partial^2\psi_1}{\partial\hat{r}^{\,2}} + \frac{1}{\hat{r}}\,\frac{\partial\psi_1}{\partial\hat{r}} - \frac{m^2}{\hat{r}^{\,2}}\,\psi_1 - \frac{J_\varphi'\,\psi_1}{\hat{r}\,(1/q - n/m)} = 0, \tag{6}$$

where

$$\hat{r} = \frac{r}{a}, \tag{7}$$

$$J_\varphi'(\hat{r}) = \frac{dJ_\varphi}{d\hat{r}}, \tag{8}$$

$$J_\varphi(\hat{r}) = \frac{R_0\,\mu_0\,j_\varphi}{B_\varphi}. \tag{9}$$

Let us assume that $J_\varphi(\hat{r})$ is a continuous function, and that $J_\varphi(\hat{r} > 1) = 0$. Note that Eq. (6) is singular at the rational surface whose minor radius, r_s, satisfies $q(r_s) = m/n$. The singularity is indicative of the breakdown of the equations of linearized, marginally-stable, ideal-MHD in the inner region.

B. Solution of Cylindrical Tearing Mode Equation

Suppose that the plasma is surrounded by a rigid conducting wall located at $r = b$, where $b > a$. Let $\psi_1(b,t) = \Psi_b(t)$, where $\Psi_b(t)$ is a complex constant that specifies the amplitude and phase of the magnetic perturbation to which the plasma is subject. The most general solution to the cylindrical tearing mode equation in the outer region is written

$$\psi_1(\hat{r},t) = \Psi_s(t)\,\hat{\psi}_s(\hat{r}) + \Psi_b(t)\,\hat{\psi}_b(\hat{r}), \tag{10}$$

Here, $\hat{\psi}_s(\hat{r})$ is a real continuous solution of Eq. (6) that satisfies

$$\hat{\psi}_s(0) = 0, \tag{11}$$

$$\hat{\psi}_s(\hat{r}_s) = 1, \tag{12}$$

$$\hat{\psi}_s(\hat{r} \geq \hat{b}) = 0, \tag{13}$$

where $\hat{r}_s = r_s/a$, and $\hat{b} = b/a$. Moreover, $\hat{\psi}_b(\hat{r})$ is a real continuous solution of Eq. (6) that satisfies

$$\hat{\psi}_b(\hat{r} \leq \hat{r}_s) = 0, \tag{14}$$

$$\hat{\psi}_b(\hat{b}) = 1. \tag{15}$$

Note that the complex quantity $\Psi_s(t)$ parameterizes the amplitude and phase of the helical magnetic flux driven by the magnetic perturbation at the interface between the inner and the outer regions.

C. Calculation of $\hat{\psi}_s(\hat{r})$

In the limit $0 < \hat{r} \ll 1$, the solution of Eq. (6), subject to the boundary condition (11), is

$$\hat{\psi}_s(\hat{r}) \propto \hat{r}^{\,m} + \kappa_0\,\hat{r}^{\,m+2}, \tag{16}$$

where

$$\kappa_0 = -\frac{x_0}{4\,(m+1)\,(1/q_0 - n/m)}, \tag{17}$$

and $q_0 = q(0)$, with $x_0 = \lim_{\hat{r}\to 0}(-J_\varphi'/\hat{r})$.

Let $\hat{x} = (\hat{r} - \hat{r}_s)/\hat{r}_s$. In the immediate vicinity of the rational surface, the continuous solution to Eq. (6), subject to the constraint (12), is

$$\hat{\psi}_s(\hat{x}) = 1 + \Delta_{s+}\,\hat{x} + \alpha_s\,\hat{x}\,\ln|\hat{x}| + \mathcal{O}(\hat{x}^2) \tag{18}$$

for $\hat{x} > 0$, and

$$\hat{\psi}_s(\hat{x}) = 1 + \Delta_{s-}\,\hat{x} + \alpha_s\,\hat{x}\,\ln|\hat{x}| + \mathcal{O}(\hat{x}^2) \tag{19}$$

for $\hat{x} < 0$, where

$$\alpha_s = -\left(\frac{\hat{r}\,q\,J_\varphi'}{s}\right)_{\hat{r}=\hat{r}_s}, \tag{20}$$

and $s = d\ln q/d\ln r$ is the *magnetic shear*. In particular,

$$\Delta_{s\,\pm} = \frac{\hat{r}}{\hat{\psi}_s}\,\frac{d\hat{\psi}_s}{d\hat{r}} - \alpha_s\,(1+\ln|\hat{x}|), \tag{21}$$

in the limit that $\hat{x}\to 0_\pm$.

Finally, solving Eq. (6) outside the plasma (where $J_\varphi' = 0$), subject to the boundary condition (13), yields

$$\left(\frac{\hat{r}}{\hat{\psi}_s}\,\frac{d\hat{\psi}_s}{d\hat{r}}\right)_{\hat{r}=1} = -m\left(\frac{\hat{b}^{\,2m}+1}{\hat{b}^{\,2m}-1}\right). \tag{22}$$

We can calculate $\hat{\psi}_s(\hat{r})$ by, first, launching a solution of Eq. (6) from the magnetic axis, subject to the boundary condition (16), and integrating to just inside the rational surface. Next, we launch a second solution of Eq. (6) from the plasma boundary, subject to the boundary condition (22), and integrate to just outside the rational surface. The two solutions are then rescaled subject to the constraint $\hat{\psi}_s(\hat{r}_s) = 1$. The real quantity

$$E_{ss} \equiv \left[\hat{r}\,\frac{d\hat{\psi}_s}{d\hat{r}}\right]_{\hat{r}=\hat{r}_{s-}}^{\hat{r}=\hat{r}_{s+}} = \Delta_{s+} - \Delta_{s-} \tag{23}$$

is the conventional tearing stability index.[12] It is assumed that $E_{ss} < 0$, which implies that the m/n tearing mode is intrinsically stable. Thus, any reconnected magnetic flux that develops at the rational surface is due solely to the action of the magnetic perturbation to which the plasma is subject.

D. Calculation of $\hat{\psi}_b(\hat{r})$

In order to calculate $\hat{\psi}_b(\hat{r})$, we launch a solution of Eq. (6) from just outside the rational surface, such that

$$\hat{\psi}_b(\hat{r}_s) = 0, \tag{24}$$

$$\frac{d\hat{\psi}_b(\hat{r}_s)}{d\hat{r}} = \frac{1}{\hat{r}_s}, \tag{25}$$

and then integrate it to the plasma boundary. Let

$$\alpha_b = \frac{1}{2m}\left[m\,\hat{b}^{\,m}\,\hat{\psi}_b(1) + \hat{b}^{\,m}\,\frac{d\hat{\psi}_b(1)}{d\hat{r}}\right], \tag{26}$$

$$\beta_b = \frac{1}{2m}\left[m\,\hat{b}^{-m}\,\hat{\psi}_b(1) - \hat{b}^{-m}\,\frac{d\hat{\psi}_b(1)}{d\hat{r}}\right], \tag{27}$$

$$\gamma_b = \alpha_b + \beta_b. \tag{28}$$

The boundary condition (15) is satisfied by dividing our solution by γ_b. Let us define the real quantity

$$E_{sb} \equiv \left[\hat{r}\,\frac{d\hat{\psi}_b}{d\hat{r}}\right]_{\hat{r}=\hat{r}_{s+}} = \frac{1}{\gamma_b}. \tag{29}$$

E. Asymptotic Matching

The complex quantity

$$\varDelta\varPsi_s(t) = \left[\hat{r}\,\frac{\partial\psi_1}{\partial\hat{r}}\right]_{\hat{r}=\hat{r}_{s-}}^{\hat{r}=\hat{r}_{s+}} \tag{30}$$

parameterizes the amplitude and phase of the shielding current that is induced in the inner region. Asymptotic matching between the inner and the outer regions yields

$$\varDelta\varPsi_s(t) = E_{ss}\,\varPsi_s(t) + E_{sb}\,\varPsi_b(t), \tag{31}$$

where use has been made of Eqs. (10), (23), (29), and (30).

F. Electromagnetic Torques

The flux-surface integrated poloidal and toroidal electromagnetic torques exerted on the plasma by the external magnetic perturbation are [1]

$$T_\theta(r,t) = T_{\theta\,s}(t)\,\delta(r-r_s), \tag{32}$$

$$T_\varphi(r,t) = T_{\varphi\,s}(t)\,\delta(r-r_s), \tag{33}$$

where

$$T_{\theta\,s}(t) = -\frac{2\pi^2\,R_0\,m}{\mu_0}\,\mathrm{Im}(\varDelta\varPsi_s\,\varPsi_s^*), \tag{34}$$

$$T_{\varphi\,s}(t) = \frac{2\pi^2\,R_0\,n}{\mu_0}\,\mathrm{Im}(\varDelta\varPsi_s\,\varPsi_s^*). \tag{35}$$

IV. INNER REGION

A. Introduction

The inner region is assumed to be governed by the four-field model described in Ref. 14. This model is an extension of a model introduced in Ref. 13, and is ultimately based on the four-field model of Hazeltine, et alia.[15]

B. Useful Definitions

The plasma is assumed to consist of two species. First, electrons of mass m_e, electrical charge $-e$, number density n_e, and temperature T_e. Second, ions of mass m_i, electrical charge $+e$, number density n_e, and temperature T_i. Let $p = n_e\,(T_e + T_i)$ be the total plasma pressure.

It is helpful to define $n_0 = n_e(r_s)$, $p_0 = p(r_s)$,

$$\eta_e = \left.\frac{d\ln T_e}{d\ln n_e}\right|_{r=r_s}, \tag{36}$$

$$\eta_i = \left.\frac{d\ln T_i}{d\ln n_e}\right|_{r=r_s}, \tag{37}$$

$$\iota = \left(\frac{T_e}{T_i}\right)_{r=r_s}\left(\frac{1+\eta_e}{1+\eta_i}\right), \tag{38}$$

where $n_e(r)$, $p(r)$, $T_e(r)$, and $T_i(r)$ refer to density, pressure, and temperature profiles that are unperturbed by the magnetic perturbation.

For the sake of simplicity, the perturbed electron and ion temperature profiles are assumed to be functions of the perturbed electron number density profile in the immediate vicinity of the rational surface. In other words, $T_e = T_e(n_e)$ and $T_i = T_i(n_e)$. This implies that $p = p(n_e)$. The MHD velocity, which is the velocity of a fictional MHD fluid, is defined as $\mathbf{V} = \mathbf{V}_E + V_{\parallel\,i}\,\mathbf{b}$, where $\mathbf{V}_E$ is the E-cross-B drift velocity, $V_{\parallel\,i}$ is the parallel component of the ion fluid velocity, and $\mathbf{b} = \mathbf{B}/|\mathbf{B}|$.

C. Dimensionless Fields

The four dimensionless fields in our four-field model—namely, ψ, ϕ, N, and V—have the following definitions:

$$\nabla\psi = \frac{\mathbf{n}\times\mathbf{B}}{r_s\,B_\varphi}, \tag{39}$$

$$\nabla\phi = \frac{\mathbf{n}\times\mathbf{V}}{r_s\,V_A}, \tag{40}$$

$$N = -\hat{d}_i\left(\frac{p-p_0}{B_\varphi^2/\mu_0}\right), \tag{41}$$

$$V = \hat{d}_i\left(\frac{\mathbf{n}\cdot\mathbf{V}}{V_A}\right). \tag{42}$$

Here, $\mathbf{n} = (0,\,\epsilon/q_s,\,1)$, $\epsilon = r/R_0$, $q_s = m/n$, $V_A = B_\varphi/\sqrt{\mu_0\,n_0\,m_i}$ is the Alfvén speed, $d_i = \sqrt{m_i/(n_0\,e^2\,\mu_0)}$ is the collisionless ion skin-depth, and $\hat{d}_i = d_i/r_s$. Our model also employs the auxiliary dimensionless field

$$J = -\frac{2\,\epsilon_s}{q_s} + \hat{\nabla}^2\psi, \tag{43}$$

where $\epsilon_s = r_s/R_0$, and $\hat{\nabla} = r_s\,\nabla$. Note that

$$\psi(r,\zeta,t) = \frac{\psi_0(r) + \psi_1(r,t)\,\mathrm{e}^{\,\mathrm{i}\,\zeta}}{r_s\,B_\varphi}, \tag{44}$$

where

$$\psi_0(r) = \frac{B_\varphi}{R_0}\int_{r_s}^{r} r'\left[\frac{n}{m} - \frac{1}{q(r')}\right]dr', \tag{45}$$

and $\psi_1(r,t)$ is defined in Eqs. (2)–(5).

D. Four-Field Model

Our four-field model takes the form:[13,14,16]

$$\frac{\partial\psi}{\partial\hat{t}} = [\phi,\psi] - \iota_e\,[N,\psi] + \hat{\eta}_\parallel\,J + \hat{E}_\parallel, \tag{46}$$

$$\frac{\partial\hat{\nabla}^2\phi}{\partial\hat{t}} = [\phi,\hat{\nabla}^2\phi] - \frac{\iota_i}{2}\left(\hat{\nabla}^2[\phi,N] + [\hat{\nabla}^2\phi,N] + [\hat{\nabla}^2 N,\phi]\right) + [J,\psi]$$
$$+\,\hat{\chi}_\varphi\,\hat{\nabla}^4(\phi + \iota_i\,N)\,, \tag{47}$$

$$\frac{\partial N}{\partial\hat{t}} = [\phi,N] + c_\beta^2\,[V,\psi] + \hat{d}_\beta^2\,[J,\psi] + \hat{\chi}_\perp\,\hat{\nabla}^2 N, \tag{48}$$

$$\frac{\partial V}{\partial \hat{t}} = [\phi, V] + [N, \psi] + \hat{\chi}_\varphi\, \hat{\nabla}^2 V. \tag{49}$$

Here, $[A, B] \equiv \hat{\nabla} A \times \hat{\nabla} B \cdot \mathbf{n}$, $\iota_e = \iota/(1+\iota)$, $\iota_i = 1/(1+\iota)$, $\hat{t} = t/(r_s/V_A)$, $\hat{\eta}_\parallel = \eta_\parallel/(\mu_0\, r_s\, V_A)$, $\hat{E}_\parallel = E_\parallel/(B_\varphi\, V_A)$, $\hat{\chi}_\varphi = \chi_\varphi/(r_s\, V_A)$, $\hat{\chi}_\perp = \chi_\perp/(r_s\, V_A)$, where $\eta_\parallel$ is the parallel plasma electrical resistivity at the rational surface, $E_\parallel$ the parallel inductive electric field that maintains the equilibrium toroidal plasma current in the vicinity of the rational surface, χ_φ the anomalous perpendicular ion momentum diffusivity at the rational surface, and $\chi_\perp$ the anomalous perpendicular energy diffusivity at the rational surface. Moreover, $d_\beta = c_\beta\, d_i$, and $\hat{d}_\beta = d_\beta/r_s$, where $c_\beta = \sqrt{\beta/(1+\beta)}$, and $\beta = (5/3)\,\mu_0\, p_0/B_\varphi^2$. Here, d_β is usually referred to as the *ion sound radius.* Note that we have neglected parallel energy transport in Eq. (48) because we expect the radial width of the inner region to be less than the critical width above which parallel transport forces the temperature and density profiles to be magnetic flux-surface functions.[17]

E. Matching to Plasma Equilibrium

The unperturbed plasma equilibrium is such that $\mathbf{B} = (0,\, B_\theta(r),\, B_\varphi)$, $p = p(r)$, $\mathbf{V} = (0,\, V_E(r),\, V_\varphi(r))$, where $V_E(r) \simeq E_r/B_\varphi$ is the (dominant θ-component of the) E-cross-B drift velocity. Now, the inner region is assumed to have a radial thickness that is much smaller than r_s. Hence, we only need to evaluate plasma equilibrium quantities in the immediate vicinity of the rational surface. Equations (39)–(42) and (44)–(45) suggest that

$$\psi(\hat{x}) = \frac{\hat{x}^{\,2}}{2\,\hat{L}_s}, \tag{50}$$

$$\phi(\hat{x}) = -\hat{V}_E\, \hat{x}, \tag{51}$$

$$N(\hat{x}) = -\hat{V}_*\, \hat{x}, \tag{52}$$

$$V = \hat{V}_\parallel, \tag{53}$$

where $\hat{L}_s = L_s/r_s$, $L_s = R_0\, q_s/s_s$, $\hat{V}_E = V_E(r_s)/V_A$, $\hat{V}_* = V_*(r_s)/V_A$, $V_*(r) = (dp/dr)/(e\, n_0\, B_\varphi)$ is the (dominant θ-component of the) diamagnetic velocity, and $\hat{V}_\parallel = \hat{d}_i\, V_\varphi(r_s)/V_A$. Here, $s_s = s(r_s)$. We also deduce that

$$J = -\left(\frac{2}{s_s} - 1\right)\frac{1}{\hat{L}_s}, \tag{54}$$

and $\hat{E}_\parallel = (2/s_s - 1)(\hat{\eta}_\parallel/\hat{L}_s)$.

F. Linearized Layer Equations

In accordance with Eqs. (43), (44), and (50)–(54), let us write

$$\psi(\hat{x},\zeta,\hat{t}) = \frac{\hat{x}^{\,2}}{2\,\hat{L}_s} + \tilde{\psi}(\hat{x},\hat{t})\,{\rm e}^{\,{\rm i}\,\zeta}, \tag{55}$$

$$\phi(\hat{x},\zeta,\hat{t}) = -\hat{V}_E\,\hat{x} + \tilde{\phi}(\hat{x},\hat{t})\,{\rm e}^{\,{\rm i}\,\zeta}, \tag{56}$$

$$N(\hat{x},\zeta,\hat{t}) = -\hat{V}_*\,\hat{x} + \iota_e\,\tilde{N}(\hat{x},\hat{t})\,{\rm e}^{\,{\rm i}\,\zeta}, \tag{57}$$

$$V(\hat{x},\zeta,\hat{t}) = \hat{V}_\parallel + \iota_e\,\tilde{V}(\hat{x},\hat{t})\,{\rm e}^{\,{\rm i}\,\zeta}, \tag{58}$$

$$J(\hat{x},\zeta,\hat{t}) = -\left(\frac{2}{s_s}-1\right)\frac{1}{\hat{L}_s} + \hat{\nabla}^2\tilde{\psi}(\hat{x},\hat{t})\,{\rm e}^{\,{\rm i}\,\zeta}. \tag{59}$$

Substituting Eqs. (55)–(59) into Eqs. (46)–(49), and only retaining terms that are first order in perturbed quantities, we obtain the following set of linearized layer equations:

$$\tau_H\,\frac{\partial\tilde{\psi}}{\partial t} + {\rm i}\,(\omega_E+\omega_{\ast\,e})\,\tau_H\,\tilde{\psi} = -{\rm i}\,\hat{x}\,(\tilde{\phi}-\tilde{N}) + S^{-1}\,\hat{\nabla}^2\tilde{\psi}, \tag{60}$$

$$\tau_H\,\frac{\partial\hat{\nabla}^2\tilde{\phi}}{\partial t} + {\rm i}\,(\omega_E+\omega_{\ast\,i})\,\tau_H\,\hat{\nabla}^2\tilde{\phi} = -{\rm i}\,\hat{x}\,\hat{\nabla}^2\tilde{\psi} + S^{-1}\,P_\varphi\,\hat{\nabla}^4\left(\tilde{\phi}+\frac{\tilde{N}}{\iota}\right), \tag{61}$$

$$\begin{aligned}\tau_H\,\frac{\partial\tilde{N}}{\partial t} + {\rm i}\,\omega_E\,\tau_H\,\tilde{N} = {}& -{\rm i}\,\omega_{\ast\,e}\,\tau_H\,\tilde{\phi} - {\rm i}\,c_\beta^{2}\,\hat{x}\,\tilde{V} - {\rm i}\,\iota_e\,\hat{d}_\beta^{\,2}\,\hat{x}\,\hat{\nabla}^2\tilde{\psi}\\ &+ S^{-1}\,P_\perp\,\hat{\nabla}^2\tilde{N},\end{aligned} \tag{62}$$

$$\tau_H\,\frac{\partial\tilde{V}}{\partial t} + {\rm i}\,\omega_E\,\tau_H\,\tilde{V} = {\rm i}\,\omega_{\ast\,e}\,\tau_H\,\tilde{\psi} - {\rm i}\,\hat{x}\,\tilde{N} + S^{-1}\,P_\varphi\,\hat{\nabla}^2\tilde{V}. \tag{63}$$

Here, $\tau_H = L_s/(m\,V_A)$ is the *hydromagnetic* time, $\omega_E = (m/r_s)\,V_E(r_s)$ the E-cross-B frequency, $\omega_{\ast\,e} = -\iota_e\,(m/r_s)\,V_\ast(r_s)$ the *electron diamagnetic* frequency, $\omega_{\ast\,i} = \iota_i\,(m/r_s)\,V_\ast(r_s)$ the *ion diamagnetic* frequency, $S = \tau_R/\tau_H$ the *Lundquist number*, $\tau_R = \mu_0\,r_s^{\,2}/\eta_\parallel$ the *resistive diffusion time*, $\tau_\varphi = r_s^{\,2}/\chi_\varphi$ the *toroidal momentum confinement* time, and $\tau_\perp = r_s^{\,2}/\chi_\perp$ the *energy confinement* time. Furthermore, $P_\varphi = \tau_R/\tau_\varphi$ and $P_\perp = \tau_R/\tau_\perp$ are *magnetic Prandtl numbers*. Note that $\iota = -\omega_{\ast\,e}/\omega_{\ast\,i}$.

In the following, in accordance with the analysis of Refs. 13 and 14, we shall neglect the term $-{\rm i}\,c_\beta^{2}\,\hat{x}\,\tilde{V}$ in Eq. (62). This approximation can be justified *a posteriori*. It is equivalent to the neglect of ion parallel dynamics in the inner region, and effectively converts our four-field model into the following three-field model:

$$\tau_H\,\frac{\partial\tilde{\psi}}{\partial t} + {\rm i}\,(\omega_E+\omega_{\ast\,e})\,\tau_H\,\tilde{\psi} = -{\rm i}\,\hat{x}\,(\tilde{\phi}-\tilde{N}) + S^{-1}\,\hat{\nabla}^2\tilde{\psi}, \tag{64}$$

$$\tau_H\,\frac{\partial\hat{\nabla}^2\tilde{\phi}}{\partial t}+{\rm i}\,(\omega_E+\omega_{\ast\,i})\,\tau_H\,\hat{\nabla}^2\tilde{\phi}=-{\rm i}\,\hat{x}\,\hat{\nabla}^2\tilde{\psi}+S^{-1}\,P_\varphi\,\hat{\nabla}^4\left(\tilde{\phi}+\frac{\tilde{N}}{\iota}\right), \tag{65}$$

$$\tau_H\,\frac{\partial\tilde{N}}{\partial t}+{\rm i}\,\omega_E\,\tau_H\,\tilde{N}=-{\rm i}\,\omega_{\ast\,e}\,\tau_H\,\tilde{\phi}-{\rm i}\,\iota_e\,\hat{d}_\beta^{\,2}\,\hat{x}\,\hat{\nabla}^2\tilde{\psi}+S^{-1}\,P_\perp\,\hat{\nabla}^2\tilde{N}. \tag{66}$$

G. Laplace Transformed Layer Equations

Let us Laplace transform Eqs. (64)–(66) by operating on them with $\int_0^\infty(\cdots)\,{\rm e}^{-g\,t}\,dt$. Assuming that all fields are zero at $t=0$, we obtain

$$g\,\tau_H\,\bar{\psi}+{\rm i}\,(\omega_E+\omega_{\ast\,e})\,\tau_H\,\bar{\psi}=-{\rm i}\,\hat{x}\,(\bar{\phi}-\bar{N})+S^{-1}\,\hat{\nabla}^2\bar{\psi}, \tag{67}$$

$$g\,\tau_H\,\hat{\nabla}^2\bar{\phi}+{\rm i}\,(\omega_E+\omega_{\ast\,i})\,\tau_H\,\hat{\nabla}^2\,\bar{\phi}=-{\rm i}\,\hat{x}\,\hat{\nabla}^2\bar{\psi}+S^{-1}\,P_\varphi\,\hat{\nabla}^4\left(\bar{\phi}+\frac{\bar{N}}{\iota}\right), \tag{68}$$

$$g\,\tau_H\,\bar{N}+{\rm i}\,\omega_E\,\tau_H\,\bar{N}=-{\rm i}\,\omega_{\ast\,e}\,\tau_H\,\bar{\phi}-{\rm i}\,\iota_e\,\hat{d}_\beta^{\,2}\,\hat{x}\,\hat{\nabla}^2\bar{\psi}+S^{-1}\,P_\perp\,\hat{\nabla}^2\bar{N}, \tag{69}$$

where

$$\bar{\psi}(\hat{x},g)=\int_0^\infty\tilde{\psi}(\hat{x},t)\,{\rm e}^{-g\,t}\,dt, \tag{70}$$

et cetera.

H. Asymptotic Matching

Let

$$\bar{\Psi}_b(g)=\int_0^\infty\Psi_b(t)\,{\rm e}^{-g\,t}\,dt, \tag{71}$$

$$\bar{\Psi}_s(g)=\int_0^\infty\Psi_s(t)\,{\rm e}^{-g\,t}\,dt, \tag{72}$$

$$\overline{\Delta\Psi}_s(g)=\int_0^\infty\Delta\Psi_s(t)\,{\rm e}^{-g\,t}\,dt\equiv\bar{\Delta}_s(g)\,\bar{\Psi}_s(g). \tag{73}$$

It follows from Eq. (31) that

$$\bar{\Psi}_s(g)=\frac{E_{sb}\,\bar{\Psi}_b(g)}{\bar{\Delta}_s(g)-E_{ss}}. \tag{74}$$

The Laplace transformed layer equations, (67)–(69), possess a twisting parity solution,[18] such that $\bar{\psi}(-\hat{x},g)=-\bar{\psi}(\hat{x},g)$, $\bar{\phi}(-\hat{x},g)=\bar{\phi}(\hat{x},g)$, and $\bar{N}(-\hat{x},g)=\bar{N}(\hat{x},g)$. In fact, an inspection of Eqs. (67)–(69) reveals that the twisting parity solution is trivial. Furthermore, when this solution is asymptotically matched to the outer region, making use of Eqs. (10), (18), (19), and (29), we obtain

$$\bar{\psi}(\hat{x},g) = \frac{\bar{\xi}_r(g)}{L_s}\,\hat{x}, \tag{75}$$

$$\bar{\phi}(\hat{x},g) = {\rm i}\,(g+{\rm i}\,\omega_E)\,\tau_H\,\frac{\bar{\xi}_r(g)}{L_s}, \tag{76}$$

$$\bar{N}(\hat{x},g) = \omega_{\ast\,e}\,\tau_H\,\frac{\bar{\xi}_r(g)}{L_s}. \tag{77}$$

Here,

$$\bar{\xi}_r(g) = \int_0^\infty \xi_r(t)\,{\rm e}^{-g\,t}\,dt, \tag{78}$$

$$\xi_r(t) = \frac{(\varDelta_{s+}+\varDelta_{s-})\,\varPsi_s(t) + E_{sb}\,\varPsi_b(t)}{2\,r_s\,B_\varphi/L_s}, \tag{79}$$

where $\xi_r(t)\,{\rm e}^{{\rm i}\,\zeta}$ is an even (in $\hat{x}$) radial displacement of the plasma in the vicinity of the rational surface that turns out to be independent of $\hat{x}$, and is driven by the magnetic perturbation. It follows that [19]

$$\tilde{\psi}(\hat{x},t) = \frac{\xi_r(t)}{L_s}\,\hat{x}, \tag{80}$$

$$\tilde{\phi}(t) = {\rm i}\,\frac{\tau_H}{L_s}\left(\frac{d}{dt}+{\rm i}\,\omega_E\right)\xi_r(t), \tag{81}$$

$$\tilde{N}(t) = \omega_{\ast\,e}\,\tau_H\,\frac{\xi_r(t)}{L_s}. \tag{82}$$

Note that the twisting parity solution gives rise to no magnetic reconnection at the rational surface [i.e., $\psi(\hat{x}=0,\zeta,\hat{t})=0$].

The Laplace transformed layer equations, (67)–(69), also possess a non-trivial tearing parity solution,[18] such that $\bar{\psi}(-\hat{x},g) = \bar{\psi}(\hat{x},g)$, $\bar{\phi}(-\hat{x},g) = -\bar{\phi}(\hat{x},g)$, and $\bar{N}(-\hat{x},g) = -\bar{N}(\hat{x},g)$. When this solution is asymptotically matched to the outer solution, making use of Eqs. (10), (18), (19), and (39), we find that

$$\bar{\psi}(\hat{x},g) \to \left[\frac{\bar{\varPsi}_s(g)}{r_s\,B_\varphi}\right]\left[1+\bar{\varDelta}_s(g)\,\frac{|\hat{x}|}{2}\right] \tag{83}$$

at the interface between the inner and the outer regions.

I. Stretched Radial Coordinate

Let us define the stretched radial coordinate $X = S^{1/3}\,\hat{x}$. Assuming that $X\sim{\cal O}(1)$ in the resonant layer that constitutes the inner region (i.e., assuming that the thickness of the layer is roughly of order $S^{-1/3}\,r_s$), and making use of the fact that $S \gg 1$ in a conventional tokamak plasma, Eqs. (67)–(69) reduce to the following set of linear layer equations:

$$[\hat{g} + \mathrm{i}\,(Q_E + Q_e)]\,\bar{\psi} = -\mathrm{i}\,X\left(\bar{\phi} - \bar{N}\right) + \frac{\partial^2\bar{\psi}}{\partial X^2}, \tag{84}$$

$$[\hat{g} + \mathrm{i}\,(Q_E + Q_i)]\,\frac{\partial^2\bar{\phi}}{\partial X^2} = -\mathrm{i}\,X\,\frac{\partial^2\bar{\psi}}{\partial X^2} + P_\varphi\,\frac{\partial^4}{\partial X^4}\left(\bar{\phi} + \frac{\bar{N}}{\iota}\right), \tag{85}$$

$$(\hat{g} + \mathrm{i}\,Q_E)\,\bar{N} = -\mathrm{i}\,Q_e\,\bar{\phi} - \mathrm{i}\,D^2\,X\,\frac{\partial^2\bar{\psi}}{\partial X^2} + P_\perp\,\frac{\partial^2\bar{N}}{\partial X^2}, \tag{86}$$

where

$$\hat{g} = S^{\,1/3}\,g\,\tau_H, \tag{87}$$

$$Q_E = S^{\,1/3}\,\omega_E\,\tau_H, \tag{88}$$

$$Q_{e,i} = S^{\,1/3}\,\omega_{*\,e,i}\,\tau_H, \tag{89}$$

$$D = S^{\,1/3}\,\iota_e^{1/2}\,\hat{d}_\beta. \tag{90}$$

According to Eq. (83), the asymptotic behavior of the tearing-parity solutions of Eqs. (84)–(86) is such that

$$\bar{\psi}(X,\hat{g}) \rightarrow \left[\frac{\bar{\Psi}_s(g)}{r_s\,B_\varphi}\right]\left[1 + \frac{\hat{\Delta}_s(\hat{g})}{2}\,|X| + {\cal O}(X^2)\right] \tag{91}$$

as $|X| \rightarrow \infty$, where

$$\bar{\Delta}_s(g) = S^{1/3}\,\hat{\Delta}_s(\hat{g}). \tag{92}$$

It follows from Eqs. (84)–(86) that

$$\bar{\phi}(X,\hat{g}) \rightarrow \mathrm{i}\,(\hat{g} + \mathrm{i}\,Q_E)\left[\frac{\bar{\Psi}_s(g)}{r_s\,B_\varphi}\right]\left[\frac{1}{X} + \frac{\hat{\Delta}_s(\hat{g})}{2}\,\mathrm{sgn}(X) + {\cal O}(X)\right] \tag{93}$$

as $|X| \rightarrow \infty$.

J. Fourier Transformation

Equations (84)–(86) are most conveniently solved in Fourier transform space.[13] Let

$$\hat{\phi}(p,\hat{g}) = \int_{-\infty}^{\infty}\bar{\phi}(X,\hat{g})\,\mathrm{e}^{-\mathrm{i}\,p\,X}\,dX, \tag{94}$$

et cetera. The Fourier transformed linear layer equations become

$$[\hat{g} + \mathrm{i}\,(Q_E + Q_e)]\,\hat{\psi} = \frac{\partial}{\partial p}\left(\hat{\phi} - \hat{N}\right) - p^2\,\hat{\psi}, \tag{95}$$

$$[\hat{g}+\mathrm{i}\,(Q_E+Q_i)]\,p^2\,\hat{\phi}=\frac{\partial(p^2\,\hat{\psi})}{\partial p}-P_\varphi\,p^4\left(\hat{\phi}+\frac{\hat{N}}{\iota}\right),\tag{96}$$

$$(\hat{g}+\mathrm{i}\,Q_E)\,\hat{N}=-\mathrm{i}\,Q_e\,\hat{\phi}-D^2\,\frac{\partial(p^2\,\hat{\psi})}{\partial p}-P_\perp\,p^2\hat{N},\tag{97}$$

where, for a tearing parity solution, Eq. (93) yields[19]

$$\hat{\phi}(p,\hat{g})\to\pi\,(\hat{g}+\mathrm{i}\,Q_E)\left[\frac{\bar{\Psi}_s(g)}{r_s\,B_\varphi}\right]\left[\frac{\hat{\Delta}_s(\hat{g})}{\pi\,p}+1+\mathcal{O}(p)\right].\tag{98}$$

as $p\to 0$.

Let

$$Y_e(p,g)\equiv\hat{\phi}(p,\hat{g})-\hat{N}(p,\hat{g})=\pi\,[\hat{g}+\mathrm{i}\,(Q_E+Q_e)]\left[\frac{\bar{\Psi}_s(g)}{r_s\,B_\varphi}\right]\,\hat{Y}_e(p,\hat{g}).\tag{99}$$

Equations (95)–(97) can be combined to give

$$\frac{\partial}{\partial p}\left[A(p,\hat{g})\,\frac{\partial\hat{Y}_e}{\partial p}\right]-\frac{B(p,\hat{g})}{C(p,\hat{g})}\,p^2\,\hat{Y}_e=0,\tag{100}$$

where

$$A(p,\hat{g})=\frac{p^2}{\hat{g}+\mathrm{i}\,(Q_E+Q_e)+p^2},\tag{101}$$

$$B(p,\hat{g})=(\hat{g}+\mathrm{i}\,Q_E)\,[\hat{g}+\mathrm{i}\,(Q_E+Q_i)]+[\hat{g}+\mathrm{i}\,(Q_E+Q_i)]\,(P_\varphi+P_\perp)\,p^2+P_\varphi\,P_\perp\,p^4,\tag{102}$$

$$C(p,\hat{g})=\hat{g}+\mathrm{i}\,(Q_E+Q_e)+\left\{P_\perp+[\hat{g}+\mathrm{i}\,(Q_E+Q_i)]\,D^2\right\}p^2+\iota_e^{-1}\,P_\varphi\,D^2\,p^4.\tag{103}$$

Furthermore, because

$$\hat{Y}_e(p,\hat{g})=\left[\frac{\hat{g}+\mathrm{i}\,(Q_E+Q_e)}{\hat{g}+\mathrm{i}\,Q_E}\right]\hat{\phi}(p,\hat{g})\tag{104}$$

as $p\to 0$, Eqs. (98) and (99) yield the following small-p boundary condition that Eq. (100) must satisfy:[19]

$$\hat{Y}_e(p,\hat{g})\to\frac{\hat{\Delta}_s(\hat{g})}{\pi\,p}+1+\mathcal{O}(p)\tag{105}$$

as $p\to 0$. Equation (100) must also satisfy the physical boundary condition

$$\hat{Y}_e(p,\hat{g})\to 0\tag{106}$$

as $p\to\infty$.

K. Reconnected Magnetic Flux

The true reconnected magnetic flux at the rational surface is

$$\Psi_0(t) \equiv \psi_1(0,t) = r_s\, B_\varphi\, \tilde{\psi}(0,t). \tag{107}$$

Let

$$\bar{\Psi}_0(g) = \int_0^\infty \Psi_0(t)\, \mathrm{e}^{-g\,t}\, dt. \tag{108}$$

It is easily demonstrated from Eqs. (84) and (95) that [5]

$$\frac{\bar{\Psi}_0(g)}{\bar{\Psi}_s(g)} \equiv F_s(\hat{g}) = -\int_0^\infty A(p,\hat{g})\,\frac{\partial \hat{Y}_e}{\partial p}\,dp. \tag{109}$$

Thus, it follows from Eqs. (74) and (92) that

$$\bar{\Psi}_0(g) = \frac{E_{sb}\, F_s(\hat{g})\, \bar{\Psi}_b(g)}{S^{1/3}\, \hat{\Delta}_s(\hat{g}) - E_{ss}}. \tag{110}$$

V. LINEAR RESONANT RESPONSE REGIMES

A. Introduction

The aim of this section is to classify all of the different linear resonant response regimes predicted by Eqs. (100), (105), (106), and (109). The approach taken is similar to that adopted in Refs. 13 and 14, except that we need to incorporate the Laplace transform parameter, g, into the analysis. In the following, we shall assume that $|Q_E| \sim |Q_e| \sim |Q_i| \sim Q$ and $P_\varphi \sim P_\perp \sim P$.

B. Constant-ψ Limit

Suppose that there are two layers in p-space. In the small-p layer, suppose that Eq. (100) reduces to

$$\frac{\partial}{\partial p}\left[\frac{p^2}{\hat{g} + \mathrm{i}\,(Q_E + Q_e) + p^2}\,\frac{\partial \hat{Y}_e}{\partial p}\right] \simeq 0 \tag{111}$$

when $p \sim R^{1/2}$, where $R = \sqrt{\hat{g}^2 + Q^2}$. Integrating directly, making use of Eq. (105), we find that

$$\hat{Y}_e(p,g) \simeq \frac{\hat{\Delta}_s}{\pi}\left[\frac{1}{p} - \frac{p}{\hat{g} + \mathrm{i}\,(Q_E + Q_e)}\right] + 1 + \mathcal{O}(p^2) \tag{112}$$

for $p \lesssim R^{1/2}$.

In the large-p limit, for $|p| \gg \mathcal{O}(R^{1/2})$, Eq. (100) reduces to

$$\frac{\partial^2 \hat{Y}_e}{\partial p^2} - \frac{B(p,\hat{g})}{C(p,\hat{g})}\, p^2\, \hat{Y}_e \simeq 0, \tag{113}$$

with $\hat{Y}_e(p,\hat{g})$ bounded as $p\to\infty$, in accordance with Eq. (106). Asymptotic matching to the small-p layer solution, (112), yields the following small-p boundary condition that the previous equation must satisfy:

$$\hat{Y}_e(p,\hat{g}) \simeq 1 - \frac{\hat{\Delta}_s}{\pi}\,\frac{p}{\hat{g}+{\rm i}\,(Q_E+Q_e)} + \mathcal{O}(p^2) \tag{114}$$

as $p\to 0$.

In the various constant-ψ linear response regimes considered in Sect. V C, Eq. (113) reduces to an equation of the form

$$\frac{\partial^2 \hat{Y}_e}{\partial p^2} - G\, p^k\, \hat{Y}_e \simeq 0, \tag{115}$$

where k is real and non-negative, and $G(\hat{g})$ is a complex constant. Let $\hat{Y}_e(p,\hat{g}) = \sqrt{p}\, Z(q)$ and $q = \sqrt{G}\, p^j/j$, where $j = (k+2)/2$. The previous equation transforms into a modified Bessel equation of general order,

$$q^2\,\frac{d^2 Z}{dq^2} + q\,\frac{dZ}{dq} - (q^2+\nu^2)\, Z = 0, \tag{116}$$

where $\nu = 1/(k+2)$. The solution that is bounded as $q\to\infty$ has the small-q expansion [20]

$$K_\nu(q) = \frac{1}{\Gamma(1-\nu)}\left(\frac{2}{q}\right)^\nu - \frac{1}{\Gamma(1+\nu)}\left(\frac{q}{2}\right)^\nu + \mathcal{O}\left(q^{2-\nu}\right), \tag{117}$$

where $\Gamma(z)$ is a gamma function. A comparison of the previous expression with Eq. (114) reveals that

$$\hat{\Delta}_s(\hat{g}) = \frac{\nu^{2\nu-1}\,\pi\,\Gamma(1-\nu)}{\Gamma(\nu)}\,[\hat{g}+{\rm i}\,(Q_E+Q_e)]\, G^\nu. \tag{118}$$

Note, finally, that $p_\ast \sim |G|^{-\nu}$, where $p_\ast$ denotes the width of the large-p layer in p-space. This width must be larger than $R^{1/2}$ (i.e., the width of the small-p layer) in order for the constant-ψ approximation to hold. Finally, it is easily demonstrated from Eqs. (106), (109), (114), and the fact that $p_\ast \gg R^{1/2}$, that

$$F_s(\hat{g}) \simeq 1 \tag{119}$$

in a constant-ψ layer. Thus, as the name suggests, a constant-ψ layer is characterized by a helical magnetic flux at the center of the layer, $\Psi_0(t)$, that is equal to that at the edge of the layer, $\Psi_s(t)$.

C. Constant-ψ Response Regimes

Suppose that $R \gg P\,p^2$ and $D^2\,p^2 \ll 1$. It follows that $k=2$, $j=2$, $\nu=1/4$, and

$$G(\hat{g}) = \frac{(\hat{g}+{\rm i}\,Q_E)\,[\hat{g}+{\rm i}\,(Q_E+Q_i)]}{\hat{g}+{\rm i}\,(Q_E+Q_e)}. \tag{120}$$

Hence, we deduce that

$$\hat{\Delta}_s(\hat{g}) = \frac{2\pi\,\Gamma(3/4)}{\Gamma(1/4)}\,[\hat{g}+{\rm i}\,(Q_E+Q_e)]^{3/4}\,(\hat{g}+{\rm i}\,Q_E)^{1/4}\,[\hat{g}+{\rm i}\,(Q_E+Q_i)]^{1/4}. \tag{121}$$

This response regime is known as the *resistive-inertial regime* because the layer response is dominated by plasma resistivity and ion inertia.[2,12,21]. The characteristic layer width is $p_\ast \sim R^{-1/4}$, which implies that the regime is valid when $P \ll R^{3/2}$, $R \gg D^4$, and $R \ll 1$.

Suppose that $R \ll P\,p^2$ and $D^2\,p^2 \ll 1$. It follows that $k=4$, $j=3$, $\nu=1/6$, and $G(\hat{g}) = P_\varphi$. Hence, we deduce that

$$\hat{\Delta}_s(\hat{g}) = \frac{6^{2/3}\,\pi\,\Gamma(5/6)}{\Gamma(1/6)}\,[\hat{g}+{\rm i}\,(Q_E+Q_e)]\,P_\varphi^{1/6}. \tag{122}$$

This response regime is known as the *viscous-resistive regime* because the layer response is dominated by ion perpendicular viscosity and plasma resistivity.[1,22] The characteristic layer width is $p_\ast \sim P^{-1/6}$, which implies that the regime is valid when $P \gg R^{3/2}$, $P \gg D^6$, and $P \ll R^{-3}$.

Suppose that $R \gg P\,p^2$ and $D^2\,p^2 \gg 1$. It follows that $k=0$, $j=1$, $\nu=1/2$, and

$$G(\hat{g}) = \frac{\hat{g}+{\rm i}\,Q_E}{D^2}. \tag{123}$$

Hence, we deduce that

$$\hat{\Delta}_s(\hat{g}) = \frac{\pi\,[\hat{g}+{\rm i}\,(Q_E+Q_e)]\,(\hat{g}+{\rm i}\,Q_E)^{1/2}}{D}. \tag{124}$$

This response regime is known as the *semi-collisional regime*.[23,24] The characteristic layer width is $p_\ast \sim D\,R^{-1/2}$, which implies that the regime is valid when $R \ll D^4$, $P \ll R^2/D^2$, and $R \ll D$.

Suppose, finally, that $R \ll P\,p^2$ and $D^2\,p^2 \gg 1$. It follows that $k=2$, $j=2$, $\nu=1/4$, and

$$G(\hat{g}) = \frac{\iota_e\,P_\perp}{D^2}. \tag{125}$$

Hence, we deduce that

$$\hat{\Delta}_s(\hat{g}) = \frac{2\pi\,\Gamma(3/4)}{\Gamma(1/4)}\,\frac{[\hat{g}+{\rm i}\,(Q_E+Q_e)]\,\iota_e^{1/4}\,P_\perp^{1/4}}{D^{1/2}}. \tag{126}$$

This response regime is known as the *diffusive-resistive regime* because the layer response is dominated by perpendicular energy diffusivity and plasma resistivity.[14] The characteristic layer width is $p_\ast \sim P^{-1/4}\,D^{1/2}$, which implies that the regime is valid when $P \gg R^2/D^2$, $P \ll D^6$, and $P \ll D^2/R^2$.

D. Non-constant-ψ Limit

Suppose that $p \ll R^{1/2}$. In this limit, Eq. (100) reduces to

$$\frac{\partial}{\partial p}\left(p^2\,\frac{\partial \hat{Y}_e}{\partial p}\right) - [\hat{g}+{\rm i}\,(Q_E+Q_e)]\,\frac{B(p,\hat{g})}{C(p,\hat{g})}\,p^2\,\hat{Y}_e = 0. \tag{127}$$

In the various non-constant-ψ regimes considered in Section V E, the previous equation takes the form

$$\frac{\partial}{\partial p}\left(p^2\,\frac{\partial \hat{Y}_e}{\partial p}\right) - G\,p^{k+2}\,\hat{Y}_e = 0, \tag{128}$$

where k is real and non-negative, and $G(\hat{g})$ is a complex constant. Let $V(p,\hat{g}) = p\,\hat{Y}_e(p,\hat{g})$. The previous equation yields

$$\frac{\partial^2 V}{\partial p^2} - G\,p^k\,V = 0. \tag{129}$$

This equation is identical in form to Equation (115), which we have already solved. Indeed, the solution that is bounded as $p\rightarrow\infty$ has the small-p expansion (117), where $q = \sqrt{G}\,p^j/j$, $j=(k+2)/2$, and $\nu = 1/(k+2)$. Matching to Equation (105) yields

$$\hat{\Delta}_s(\hat{g}) = -\frac{\pi\,\nu^{1-2\nu}\,\Gamma(\nu)}{\Gamma(1-\nu)}\,G^{-\nu}. \tag{130}$$

The layer width in p-space again scales as $p_\ast \sim |G|^{-\nu}$. This width must be less that $R^{1/2}$. Finally, it follows from Eqs. (101) and (109) that

$$F_s(\hat{g}) = \frac{2}{\hat{g}+{\rm i}\,(Q_E+Q_e)}\int_0^\infty V(p,\hat{g})\,dp \tag{131}$$

in a non-constant-ψ response regime.

E. Non-constant-ψ Linear Resonant Response Regimes

Suppose that $R \gg P\,p^2$ and $D^2\,p^2 \ll 1$. It follows that $k=0$, $j=1$, $\nu=1/2$,

$$G(\hat{g}) = (\hat{g} + {\rm i}\,Q_E)\,[\hat{g} + {\rm i}\,(Q_E+Q_i)], \tag{132}$$

$$V(p,\hat{g}) = -G^{-1/2}\,\exp(-G^{1/2}\,p). \tag{133}$$

Hence, we deduce that

$$\hat{\Delta}_s(\hat{g}) = -\frac{\pi}{(\hat{g}+{\rm i}\,Q_E)^{1/2}\,[\hat{g}+{\rm i}\,(Q_E+Q_i)]^{1/2}}, \tag{134}$$

$$F_s(\hat{g}) = -\frac{2}{[\hat{g}+{\rm i}\,(Q_E+Q_e)]\,(\hat{g}+{\rm i}\,Q_E)\,[\hat{g}+{\rm i}\,(Q_E+Q_i)]}. \tag{135}$$

This response regime is known as the *inertial regime* because the layer response is dominated by ion inertia.[2,21] The characteristic layer width is $p_\ast \sim R^{-1}$, which implies that the regime is valid when $P \ll R^3$, $R \gg D$, and $R \gg 1$.

Suppose that $R \ll P\,p^2$ and $D^2\,p^2 \ll 1$. It follows that $k=2$, $j=2$, $\nu=1/4$, and

$$G(\hat{g}) = [\hat{g}+{\rm i}\,(Q_E+Q_e)]\,P_\varphi, \tag{136}$$

$$V(p,\hat{g}) = -\frac{\Gamma(1/4)}{\sqrt{\pi}\,2^{3/4}\,G^{1/4}}\,U(0,2^{1/2}\,G^{1/4}\,p), \tag{137}$$

where $U(a,x) = D_{-a-1/2}(x)$ is a parabolic cylinder function.[20] Hence, we deduce that

$$\hat{\Delta}_s(\hat{g}) = -\frac{\pi}{2}\,\frac{\Gamma(1/4)}{\Gamma(3/4)}\,\frac{1}{[\hat{g}-{\rm i}\,(Q_E+Q_e)]^{1/4}\,P_\varphi^{1/4}}, \tag{138}$$

$$F_s(\hat{g}) = -\frac{\pi^{1/2}}{2}\,\frac{\Gamma(1/4)}{\Gamma(3/4)}\,\frac{1}{[\hat{g}-{\rm i}\,(Q_E+Q_e)]^{3/2}\,P_\varphi^{1/2}}, \tag{139}$$

where use has been made of the result

$$\int_0^\infty U(0,x)\,dx = \frac{\pi}{2^{3/4}\,\Gamma(3/4)}. \tag{140}$$

This response regime is known as the *viscous-inertial regime* because the layer response is dominated by ion perpendicular viscosity and ion inertia.[2] The characteristic layer width is $p_\ast \sim R^{-1/4}\,P^{-1/4}$, which implies that the regime is valid when $P \gg R^3$, $P \gg D^4/R$, and $P \gg R^{-3}$.

Suppose, finally, that $R \ll P\,p^2$ and $D^2\,p^2 \gg 1$. It follows that $k=0$, $j=1$, $\nu=1/2$, and

$$G(\hat{g}) = \frac{[\hat{g}+{\rm i}\,(Q_E+Q_e)]\,\iota_e\,P_\perp}{D^2}, \tag{141}$$

$$V(p,\hat{g}) = -G^{-1/2}\,\exp(-G^{1/2}\,p). \tag{142}$$

Hence, we deduce that

$$\hat{\Delta}_s = -\frac{\pi\,D}{[\hat{g}+{\rm i}\,(Q_E+Q_e)]^{1/2}\,\iota_e^{1/2}\,P_\perp^{1/2}}, \tag{143}$$

$$F_s(\hat{g}) = -\frac{2\,D^2}{[\hat{g}+{\rm i}\,(Q_E+Q_e)]^2\,\iota_e\,P_\perp}. \tag{144}$$

This response regime is known as the *diffusive-inertial regime* because the layer response is dominated by perpendicular energy diffusivity and ion inertia.[14] The characteristic layer width is $p_\ast \sim R^{-1/2}\,P^{-1/2}\,D$, which implies that the regime is valid when $R \ll D$, $P \ll D^4/Q$, and $P \gg D^2/R^2$.

F. Linear Resonant Response Regimes

Figures 1 and 2 show the extents of the various different linear resonant response regimes in R-P space for the cases $D<1$ and $D>1$, respectively. Note that Fig. 2 differs slightly from Fig. 2 in Ref. 14 because we are now demanding equal layer thicknesses at the boundaries between the various response regimes.

VI. LINEAR RESPONSE THEORY

A. Introduction

The aim of this section is to calculate the *linear* response of the resonant layer to the resonant magnetic perturbation.

B. Model Resonant Magnetic Perturbation

Suppose that the resonant magnetic perturbation takes the form

$$\Psi_b(t) = \Psi_{b\,0}\left\{\begin{array}{ll} 0 & \quad t<0\\[1ex] 1 & \quad t\geq 0\end{array}\right.. \tag{145}$$

Equations (71) and (145) yield[19]

$$\bar{\Psi}_b(g) = \Psi_{b\,0}\left(\frac{1}{g+\gamma_g}-\frac{1}{g}\right). \tag{146}$$

Thus, it follows from Eqs. (73), (74), (92), and (110) that

$$\overline{\Delta\Psi}_s(g) = E_{sb}\,\Psi_{b\,0}\left[\frac{S^{1/3}\,\hat{\Delta}_s(\hat{g})}{S^{1/3}\,\hat{\Delta}_s(\hat{g})-E_{ss}}\right]\frac{1}{g}, \tag{147}$$

$$\bar{\Psi}_0(g) = E_{sb}\,\Psi_{b\,0}\left[\frac{F_s(\hat{g})}{S^{1/3}\,\hat{\Delta}_s(\hat{g})-E_{ss}}\right]\frac{1}{g}. \tag{148}$$

Recall that $\Delta\Psi_s(t)$ [which is the inverse Laplace transform of $\overline{\Delta\Psi}_s(g)$] is a measure of the net shielding current that develops in the resonant layer, in response to the magnetic perturbation. On the other hand, $\Psi_0(t)$ [which is the inverse Laplace transform of $\bar{\Psi}_0(g)$] is the reconnected magnetic flux driven at the rational surface by the magnetic perturbation.

Under normal circumstances, we expect $S^{1/3} \gg 1$, and $|E_{ss}|$, $|\hat{\Delta}_s| \sim \mathcal{O}(1)$. Thus, Eqs. (147) and (148) simplify to give

$$\overline{\Delta\Psi}_s(g) \simeq E_{sb}\,\Psi_{b\,0}\left(\frac{1}{g+\gamma_g}-\frac{1}{g}\right), \tag{149}$$

$$\bar{\Psi}_0(g) \simeq E_{sb}\,\Psi_{b\,0}\left[\frac{F_s(\hat{g})}{S^{1/3}\,\hat{\Delta}_s(\hat{g})}\right]\left(\frac{1}{g+\gamma_g}-\frac{1}{g}\right). \tag{150}$$

Equation (149) can be inverse Laplace transformed to yield

$$\Delta\Psi_s(t) = E_{sb}\,\Psi_{b\,0}\left(\mathrm{e}^{\gamma_g\,t}-1\right) = E_{sb}\,\Psi_b(t). \tag{151}$$

According to the previous equation, the net shielding current that develops in the resonant layer is directly proportional to the amplitude of the magnetic perturbation, and is independent of the layer physics. This implies that the net shielding current grows exponentially in time at the same rate as the magnetic perturbation.

C. Layer Widths

Let δ be the radial width of the resonant layer, and let $\hat{\delta} = \delta/r_s$. According to the analysis of Sects. V B and V D, $\hat{\delta} \simeq S^{-1/3}\,|G|^{\nu}$. In estimating the layer width, we shall assume that at time t the dominant layer response is associated with the following value of the Laplace transform variable:

$$g = \frac{1}{t}\left(1+\gamma_g^{\,2}\,t^2\right)^{1/2}. \tag{152}$$

Thus, $g \simeq 1/t$ for $\gamma_g\,t \ll 1$, and $g \simeq \gamma_g$ for $\gamma_g\,t \gg 1$.

In the inertial response regime, we find that

$$\hat{\delta}_I(t) = \frac{\tau_H}{t}\left[1+(\gamma_g^{\,2}+\omega_E^{\,2})\,t^2\right]^{1/4}\left(1+[\gamma_g^{\,2}+(\omega_E+\omega_{\ast\,i})^2]\,t^2\right)^{1/4}. \tag{153}$$

Thus, at small times, the layer width, which is also the width of the shielding current, decreases in time as t^{-1}. However, this decrease eventually stalls, and the layer width attains a constant value.

In the viscous-inertial response regime, we find that

$$\hat{\delta}_{VI}(t) = \left(\frac{\tau_H}{\tau_\varphi}\right)^{1/4}\left(\frac{\tau_H}{t}\right)^{1/4}\left(1+[\gamma_g^{\,2}+(\omega_E+\omega_{\ast\,e})^2]\,t^2\right)^{1/8}. \tag{154}$$

Thus, at small times, the layer width decreases in time as $t^{-1/4}$. However, this decrease eventually stalls, and the layer width attains a constant value.

In the diffusive-inertial response regime, we find that

$$\hat{\delta}_{DI}(t) = \frac{1}{\hat{d}_\beta}\left(\frac{\tau_H}{\tau_\perp}\right)^{1/2}\left(\frac{\tau_H}{t}\right)^{1/2}\left(1+[\gamma_g^{\,2}+(\omega_E+\omega_{\ast\,e})^2]\,t^2\right)^{1/4} \tag{155}$$

Thus, at small times, the layer width decreases in time as $t^{-1/2}$. However, this decrease eventually stalls, and the layer width attains a constant value.

In the semi-collisional response regime, we find that

$$\hat{\delta}_{SC}(t) = \frac{1}{\hat{d}_\beta}\left(\frac{\tau_H}{\tau_R}\right)^{1/2}\left(\frac{\tau_H}{t}\right)^{1/2}\left[1+(\gamma_g^{\,2}+\omega_E^{\,2})\,t^2\right]^{1/4}. \tag{156}$$

Thus, at small times, the layer width decreases in time as $t^{-1/2}$. However, this decrease eventually stalls, and the layer width attains a constant value.

In the resistive-inertial response regime, we find that

$$\hat{\delta}_{RI}(t) = \left(\frac{\tau_H}{\tau_R}\right)^{1/4}\left(\frac{\tau_H}{t}\right)^{1/4}\frac{\left[1+(\gamma_g^{\,2}+\omega_E^{\,2})\,t^2\right]^{1/8}\left(1+[\gamma_g^{\,2}+(\omega_E+\omega_{\ast\,i})^2]\,t^2\right)^{1/8}}{\left(1+[\gamma_g^{\,2}+(\omega_E+\omega_{\ast\,e})^2]\,t^2\right)^{1/8}}. \tag{157}$$

Thus, at small times, the layer width decreases in time as $t^{-1/4}$. However, this decrease eventually stalls, and the layer width attains a constant value.

In the viscous-resistive response regime, we find that

$$\hat{\delta}_{VR} = \frac{\tau_H^{\,1/3}}{\tau_R^{\,1/6}\,\tau_\varphi^{\,1/6}} \tag{158}$$

Thus, the layer width is independent of time.

Finally, in the diffusive-resistive response regime, we find that

$$\hat{\delta}_{DR} = \frac{1}{\hat{d}_\beta^{1/2}}\,\frac{\tau_H^{1/2}}{\tau_R^{1/4}\,\tau_\perp^{1/4}}. \tag{159}$$

Thus, the layer width is independent of time.

D. Reconnection Scenario 1

The layer widths calculated in Sect. VI C allow us to determine how the plasma response in the resonant layer navigates through the various linear response regimes. This navigation is illustrated in Figs. 3 and 4. Here, $\hat{t} = t/(S^{1/3}\,\tau_H)$.

Consider reconnection scenario 1, which corresponds to the horizontal lines labelled 1 in Figs. 3 and 4. This scenario holds when $D < 1$ and $P > 1$, or when $D > 1$ and $P > D^6$.

The plasma in the resonant layer is initially in the inertial response regime. It follows from Eqs. (134), (135), and (150) that

$$\frac{\bar{\Psi}_0(g)}{E_{sb}\,\Psi_{b\,0}} = \frac{2}{\pi}\,\frac{1}{\tau_H\,\tau_R}\,\frac{1}{[g+\mathrm{i}\,(\omega_E+\omega_{*\,e})]\,(g+\mathrm{i}\,\omega_E)^{1/2}\,[g+\mathrm{i}\,(\omega_E+\omega_{*\,i})]^{1/2}} \times\left(\frac{1}{g+\gamma_g}-\frac{1}{g}\right), \tag{160}$$

which can be inverse Laplace transformed to give [19]

$$\frac{\Psi_0(t)}{E_{sb}\,\Psi_{b\,0}} = \frac{2}{\pi}\,\frac{1}{\tau_H\,\tau_R}\int_0^t F_I(t,t')\,\mathrm{e}^{-\mathrm{i}\,(\omega_E+\omega_{*\,i}/2)\,t'}\,J_0\!\left(\frac{\omega_{*\,i}\,t'}{2}\right)dt', \tag{161}$$

where

$$F_I(t,t') = \frac{\mathrm{e}^{\gamma_g\,(t-t')}}{\gamma_g+\mathrm{i}\,(\omega_E+\omega_{*\,e})} - \frac{1}{\mathrm{i}\,(\omega_E+\omega_{*\,e})} + \frac{\gamma_g\,\mathrm{e}^{-\mathrm{i}\,(\omega_E+\omega_{*\,e})\,(t-t')}}{\mathrm{i}\,(\omega_E+\omega_{*\,e})\,[\gamma_g+\mathrm{i}\,(\omega_E+\omega_{*\,e})]}. \tag{162}$$

Here, $J_\nu(x)$ is a Bessel function of general order ν.[20] In the small-t limit, $\gamma_g\,t$, $|\omega_{*\,e}|\,t$, $|\omega_{*\,i}|\,t$, $|\omega_E|\,t \ll 1$, Eq. (161) reduces to

$$\frac{\Psi_0(t)}{E_{sb}\,\Psi_{b\,0}} = \frac{1}{3\pi}\,\frac{\gamma_g\,t^3}{\tau_H\,\tau_R}. \tag{163}$$

In other words, the reconnected magnetic flux initially grows in time as t^3. In the large-t limit, $\gamma_g\,t$, $|\omega_{*\,e}|\,t$, $|\omega_{*\,i}|\,t$, $|\omega_E|\,t \gg 1$, Eq. (161) reduces to [25]

$$\frac{\Psi_0(t)}{E_{sb}\,\Psi_{b\,0}} = \frac{\mathrm{e}^{\gamma_g\,t}}{\Sigma_I}, \tag{164}$$

where

$$\Sigma_I = \frac{\pi}{2}\,\tau_H\,\tau_R\,[\gamma_g + {\rm i}\,(\omega_E+\omega_{\ast\,e})]\,(\gamma_g+{\rm i}\,\omega_E)^{1/2}\,[\gamma_g+{\rm i}\,(\omega_E+\omega_{\ast\,i})]^{1/2}. \tag{165}$$

Note that expression (164) corresponds to the contribution of the pole at $g = -\gamma_g$ when expression (160) is inverse Laplace transformed. According to Eq. (164), at large t, the reconnected magnetic flux grows exponentially in time at the same rate as the magnetic perturbation. The parameter $\Sigma_I \sim {\cal O}(S^{1/3})$ is termed the *shielding factor*, and is, roughly speaking, the factor by which the driven reconnected magnetic flux is reduced by the shielding current, relative to that in the absence of shielding.

In order for the plasma in the resonant layer to lie in the inertial, rather than the viscous-inertial, response regime, we require $\hat{\delta}_I > \hat{\delta}_{VI}$. In other words, the inertial layer width must exceed the viscous-inertial layer width. According to Eqs. (153) and (154), the plasma lies in the inertial response regime provided that

$$t < \tau_{I-VI}\,\frac{\left[1+(\gamma_g^{\,2}+\omega_E^{\,2})\,t^2\right]^{1/3}\left(1+[\gamma_g^{\,2}+(\omega_E+\omega_{\ast\,i})^2]\,t^2\right)^{1/3}}{\left(1+[\gamma_g^{\,2}+(\omega_E+\omega_{\ast\,e})^2]\,t^2\right)^{1/6}}, \tag{166}$$

where

$$\tau_{I-VI} = \tau_H^{\,2/3}\,\tau_\varphi^{\,1/3}. \tag{167}$$

Let us assume that $\omega_{\ast\,e}\sim\omega_{\ast\,i}\sim\omega_E\sim\omega_\ast$, where $\omega_\ast = -(m/r_s)\,V_\ast(r_s)$ is the diamagnetic frequency. Let $\Omega_{g\ast} = \sqrt{\gamma_g^{\,2}+\omega_\ast^{\,2}}$. There are two possibilities. If $\Omega_{g\ast}\,\tau_{I-VI}\ll 1$ then the plasma in the resonant layer leaves the inertial response regime when $t = \tau_{I-VI}$, and enters the viscous-inertial response regime. Moreover, the plasma leaves the inertial regime long before the large-t state (164) is achieved. On the other hand, if $\Omega_{g\ast}\,\tau_{I-VI}\gg 1$ then the plasma in the resonant layer never leaves the inertial response regime, and the large-t state (164) is eventually achieved. In the latter case, the trajectories labelled 1 in Figs. 3 and 4 terminate in the inertial response regime. In the former case, the trajectories enter the viscous-inertial response regime.

In the viscous-inertial response regime, Eqs. (138), (139), and (150) yield

$$\frac{\bar{\Psi}_0(g)}{E_{sb}\,\Psi_{b\,0}} = \frac{1}{\pi^{1/2}}\,\frac{1}{\tau_H^{1/2}\,\tau_R\,\tau_\varphi^{-1/4}}\,\frac{1}{[g+{\rm i}\,(\omega_E+\omega_{\ast\,e})]^{5/4}}\left(\frac{1}{g+\gamma_g}-\frac{1}{g}\right), \tag{168}$$

which can be inverse Laplace transformed to give [19]

$$\frac{\Psi_0(t)}{E_{sb}\,\Psi_{b\,0}} = \frac{1}{\pi^{1/2}\,\Gamma(5/4)}\,\frac{1}{\tau_H^{1/2}\,\tau_R\,\tau_\varphi^{-1/4}}\int_0^t\left[{\rm e}^{\gamma_g\,(t-t')}-1\right]t'^{\,1/4}\,{\rm e}^{-{\rm i}\,(\omega_E+\omega_{\ast\,e})\,t'}\,dt'. \tag{169}$$

In the small-t limit, $\gamma_g\,t$, $|\omega_{\ast\,e}|\,t$, $|\omega_E|\,t \ll 1$, Eq. (169) reduces to

$$\frac{\Psi_0(t)}{E_{sb}\,\Psi_{b\,0}} = \frac{16}{45\,\pi^{1/2}\,\Gamma(5/4)}\,\frac{\gamma_g\,t^{9/4}}{\tau_H^{1/2}\,\tau_R\,\tau_\varphi^{-1/4}}. \tag{170}$$

In other words, the reconnected magnetic flux initially grows in time as $t^{9/4}$. In the large-t limit, $\gamma_g\,t$, $|\omega_{\ast\,e}|\,t$, $|\omega_E|\,t \gg 1$, Eq. (169) reduces to [25]

$$\frac{\Psi_0(t)}{E_{sb}\,\Psi_{b\,0}} = \frac{{\rm e}^{\gamma_g\,t}}{\Sigma_{VI}}, \tag{171}$$

where

$$\Sigma_{VI} = \pi^{1/2}\,\tau_H^{1/2}\,\tau_R\,\tau_\varphi^{-1/4}\,[\gamma_g + {\rm i}\,(\omega_E+\omega_{\ast\,e})]^{5/4}. \tag{172}$$

Expression (171) corresponds to the contribution of the pole at $g = -\gamma_g$ when expression (168) is inverse Laplace transformed. Moreover, $\Sigma_{VI} \sim {\cal O}(S^{1/3})$ is the shielding factor.

In order for the plasma in the resonant layer to lie in the viscous-inertial, rather than the viscous-resistive, response regime, we require $\hat{\delta}_{VI} > \hat{\delta}_{VR}$. According to Eqs. (154) and (158), the plasma lies in the viscous-inertial response regime provided that

$$t < \tau_{VI-VR}\,\left(1 + [\gamma_g^{\,2} + (\omega_E+\omega_{\ast\,e})^2]\,t^2\right)^{1/2}, \tag{173}$$

where

$$\tau_{VI-VR} = \frac{\tau_H^{2/3}\,\tau_R^{2/3}}{\tau_\varphi^{1/3}}. \tag{174}$$

There are two possibilities. If $\Omega_{g\ast}\,\tau_{VI-VR} \ll 1$ then the plasma in the resonant layer leaves the viscous-inertial response regime when $t = \tau_{VI-VR}$, and enters the viscous-resistive response regime. Moreover, the plasma leaves the viscous-inertial regime long before the large-t state (171) is achieved. On the other hand, if $\Omega_{g\ast}\,\tau_{VI-VR} \gg 1$ then the plasma in the resonant layer never leaves the viscous-inertial response regime, and the large-t state (171) is eventually achieved. In the latter case, the trajectories labelled 1 in Figs. 3 and 4 terminate in the viscous-inertial response regime. In the former case, the trajectories enter the viscous-resistive response regime.

In the viscous-resistive response regime, Eqs. (119), (122), and (150) yield

$$\frac{\bar{\Psi}_0(g)}{E_{sb}\,\Psi_{b\,0}} = \frac{\Gamma(1/6)}{6^{2/3}\,\pi\,\Gamma(5/6)}\,\frac{1}{\tau_H^{2/3}\,\tau_R^{1/2}\,\tau_\varphi^{-1/6}}\,\frac{1}{g+{\rm i}\,(\omega_E+\omega_{\ast\,e})}\left(\frac{1}{g+\gamma_g}-\frac{1}{g}\right), \tag{175}$$

which can be inverse Laplace transformed to give [19]

$$\frac{\Psi_0(t)}{E_{sb}\,\Psi_{b\,0}} = \frac{\Gamma(1/6)}{6^{2/3}\,\pi\,\Gamma(5/6)}\,\frac{1}{\tau_H^{2/3}\,\tau_R^{1/2}\,\tau_\varphi^{-1/6}}\left[\frac{{\rm e}^{\gamma_g\,t}-{\rm e}^{-{\rm i}\,(\omega_E+\omega_{\ast\,e})\,t}}{\gamma_g+{\rm i}\,(\omega_E+\omega_{\ast\,e})} - \frac{1-{\rm e}^{-{\rm i}\,(\omega_E+\omega_{\ast\,e})\,t}}{{\rm i}\,(\omega_E+\omega_{\ast\,e})}\right]. \tag{176}$$

In the small-t limit, $\gamma_g\,t$, $|\omega_{\ast\,e}|\,t$, $|\omega_E|\,t\ll 1$, Eq. (176) reduces to

$$\frac{\Psi_0(t)}{E_{sb}\,\Psi_{b\,0}}=\frac{\Gamma(1/6)}{6^{2/3}\,2\pi\,\Gamma(5/6)}\,\frac{\gamma_g\,t^2}{\tau_H^{\,2/3}\,\tau_R^{\,1/2}\,\tau_\varphi^{\,-1/6}}. \tag{177}$$

In other words, the reconnected magnetic flux initially grows in time as t^2. In the large-t limit, $\gamma_g\,t$, $|\omega_{\ast\,e}|\,t$, $|\omega_E|\,t\gg 1$, Eq. (169) reduces to

$$\frac{\Psi_0(t)}{E_{sb}\,\Psi_{b\,0}}=\frac{{\rm e}^{\,\gamma_g\,t}}{\Sigma_{VR}}, \tag{178}$$

where

$$\Sigma_{VR}=\frac{6^{2/3}\,\pi\,\Gamma(5/6)}{\Gamma(1/6)}\,\tau_H^{\,2/3}\,\tau_R^{\,1/2}\,\tau_\varphi^{\,-1/6}\,[\gamma_g+{\rm i}\,(\omega_E+\omega_{\ast\,e})]. \tag{179}$$

Expression (178) corresponds to the contribution of the pole at $g=-\gamma_g$ when expression (175) is inverse Laplace transformed. Moreover, $\Sigma_{VR}\sim{\cal O}(S^{1/3})$ is the shielding factor.

E. Reconnection Scenario 2

Consider reconnection scenario 2, which corresponds to the horizontal line labelled 2 in Fig. 3. This scenario holds when $D<1$ and $D^6<P<1$.

The plasma in the resonant layer is initially in the inertial response regime. Driven magnetic reconnection in this response regime is described in Sect. VI D.

In order for the plasma in the resonant layer to lie in the inertial, rather than the resistive-inertial, response regime, we require $\hat{\delta}_I>\hat{\delta}_{RI}$. According to Eqs. (153) and (157), the plasma lies in the inertial response regime provided that

$$\begin{aligned}t<\tau_{I-RI}\left[1+(\gamma_g^{\,2}+\omega_E^{\,2})\,t^2\right]^{1/6}&\left(1+[\gamma_g^{\,2}+(\omega_E+\omega_{\ast\,i})^2]\,t^2\right)^{1/6}\\ \times\left(1+[\gamma_g^{\,2}+(\omega_E+\omega_{\ast\,e})^2]\,t^2\right)^{1/6}&,\end{aligned} \tag{180}$$

where

$$\tau_{I-RI}=\tau_H^{\,2/3}\,\tau_R^{\,1/3}. \tag{181}$$

There are two possibilities. If $\Omega_{g\ast}\,\tau_{I-RI}\ll 1$ then the plasma in the resonant layer leaves the inertial response regime when $t=\tau_{I-RI}$, and enters the resistive-inertial response regime. Moreover, the plasma leaves the inertial response regime long before the large-t state (164) is achieved. On the other hand, if $\Omega_{g\ast}\,\tau_{I-RI}\gg 1$ then the plasma in the resonant layer never leaves the inertial response regime, and the large-t state (164) is eventually achieved. In the latter case, the trajectory labelled 2 in Fig. 3 terminates in the inertial response regime. In the former case, the trajectory enters the resistive-inertial response regime.

In the resistive-inertial response regime, Eqs. (119), (121), and (150) yield

$$\frac{\bar{\Psi}_0(g)}{E_{sb}\,\Psi_{b\,0}} = \frac{\Gamma(1/4)}{2\pi\,\Gamma(3/4)}\,\frac{1}{\tau_H^{1/2}\,\tau_R^{3/4}}\,\frac{1}{[g+{\rm i}\,(\omega_E+\omega_{\ast\,e})]^{3/4}\,(g+{\rm i}\,\omega_E)^{1/4}\,[g+{\rm i}\,(\omega_E+\omega_{\ast\,i})]^{1/4}}$$
$$\times\left(\frac{1}{g+\gamma_g}-\frac{1}{g}\right), \tag{182}$$

which can be inverse Laplace transformed to give [19]

$$\frac{\Psi_0(t)}{E_{sb}\,\Psi_{b\,0}} = \frac{\Gamma(1/4)}{(2\pi)^{3/2}\,\Gamma(3/4)}\,\frac{1}{\tau_H^{1/2}\,\tau_R^{3/4}}\int_0^t\left[{\rm e}^{\gamma_g\,(t-t')}-1\right]\,{\rm e}^{-{\rm i}\,(\omega_E+\omega_{\ast\,e})\,t'}\,F_{RI}(t')\,dt', \tag{183}$$

where

$$F_{RI}(t') = \int_0^{t'}(t'-t'')^{-1/4}\,t''^{-1/4}\,{\rm e}^{{\rm i}\,(\omega_{\ast\,e}+\omega_{\ast\,i}/2)\,t''}\,\omega_{\ast\,i}^{1/4}\,J_{-1/4}\!\left(\frac{\omega_{\ast\,i}\,t''}{2}\right)dt''. \tag{184}$$

In the small-t limit, $\gamma_g\,t,\ |\omega_{\ast\,e}|\,t,\ |\omega_{\ast\,i}|\,t,\ |\omega_E|\,t\ll 1$, Eq. (183) reduces to

$$\frac{\Psi_0(t)}{E_{sb}\,\Psi_{b\,0}} = \frac{32}{45\pi}\,\frac{\gamma_g\,t^{9/4}}{\tau_H^{1/2}\,\tau_R^{3/4}}. \tag{185}$$

In other words, the reconnected magnetic flux initially grows in time as $t^{9/4}$. In the large-t limit, $\gamma_g\,t,\ |\omega_{\ast\,e}|\,t,\ |\omega_{\ast\,i}|\,t,\ |\omega_E|\,t\gg 1$, Eq. (183) reduces to [25]

$$\frac{\Psi_0(t)}{E_{sb}\,\Psi_{b\,0}} = \frac{{\rm e}^{\gamma_g\,t}}{\Sigma_{RI}}, \tag{186}$$

where

$$\Sigma_{RI} = \frac{2\pi\,\Gamma(3/4)}{\Gamma(1/4)}\,\tau_H^{1/2}\,\tau_R^{3/4}\,[\gamma_g+{\rm i}\,(\omega_E+\omega_{\ast\,e})]^{3/4}\,(\gamma_g+{\rm i}\,\omega_E)^{1/4}\,[\gamma_g+{\rm i}\,(\omega_E+\omega_{\ast\,i})]^{1/4}. \tag{187}$$

Expression (186) corresponds to the contribution of the pole at $g=-\gamma_g$ when expression (182) is inverse Laplace transformed. Moreover, $\Sigma_{RI}\sim{\cal O}(S^{1/3})$ is the shielding factor.

In order for the plasma in the resonant layer to lie in the resistive-inertial, rather than the viscous-resistive, response regime, we require $\hat{\delta}_{RI}>\hat{\delta}_{VR}$. According to Eqs. (157) and (158), the plasma lies in the resistive-inertial regime provided that

$$t<\tau_{RI-VR}\,\frac{\left[1+(\gamma_g^{\,2}+\omega_E^{\,2})\,t^2\right]^{1/2}\left(1+[\gamma_g^{\,2}+(\omega_E+\omega_{\ast\,i})^2]\,t^2\right)^{1/2}}{\left(1+[\gamma_g^{\,2}+(\omega_E+\omega_{\ast\,e})^2]\,t^2\right)^{1/2}}, \tag{188}$$

where

$$\tau_{RI-VR} = \frac{\tau_H^{2/3}\,\tau_\varphi^{2/3}}{\tau_R^{1/3}}. \tag{189}$$

There are two possibilities. If $\Omega_{g*}\,\tau_{RI-VR} \ll 1$ then the plasma in the resonant layer leaves the resistive-inertial response regime when $t = \tau_{RI-VR}$, and enters the viscous-resistive response regime. Moreover, the plasma leaves the inertial response regime long before the large-t state (186) is achieved. On the other hand, if $\Omega_{g*}\,\tau_{RI-VR} \gg 1$ then the plasma in the resonant layer never leaves the resistive-inertial response regime, and the large-t state (186) is eventually achieved. In the latter case, the trajectory labelled 2 in Fig. 3 terminates in the resistive-inertial response regime. In the former case, the trajectory enters the viscous-resistive response regime.

Driven magnetic reconnection in the viscous-resistive response regime is described in Sect. VI D.

F. Reconnection Scenario 3

Consider reconnection scenario 3, which corresponds to the horizontal line labelled 3 in Fig. 3. This scenario holds when $D < 1$ and $P < D^6$.

The plasma in the resonant layer is initially in the inertial response regime. Driven magnetic reconnection in this response regime is described in Sect. VI D.

If $\Omega_{g*}\,\tau_{I-RI} \ll 1$ then the plasma in the resonant layer leaves the inertial response regime when $t = \tau_{I-RI}$, and enters the resistive-inertial response regime. On the other hand, if $\Omega_{g*}\,\tau_{I-RI} \gg 1$ then the plasma in the resonant layer never leaves the inertial response regime, and the large-t state (164) is eventually achieved. In the latter case, the trajectory labelled 3 in Fig. 3 terminates in the inertial response regime. In the former case, the trajectory enters the resistive-inertial response regime.

Driven magnetic reconnection in the resistive-inertial response regime is described in Sect. VI E.

In order for the plasma in the resonant layer to lie in the resistive-inertial, rather than the semi-collisional, response regime, we require $\hat{\delta}_{SC} > \hat{\delta}_{RI}$. According to Eqs. (156) and (157), the plasma lies in the resistive-inertial response regime provided that

$$t < \tau_{RI-SC}\,\frac{\left[1+(\gamma_g^{\,2}+\omega_E^{\,2})\,t^2\right]^{1/2}\left(1+\left[\gamma_g^{\,2}+(\omega_E+\omega_{*\,i})^2\right]t^2\right)^{1/2}}{\left(1+\left[\gamma_g^{\,2}+(\omega_E+\omega_{*\,e})^2\right]t^2\right)^{1/2}}, \tag{190}$$

where

$$\tau_{RI-SC} = \frac{\tau_H^2}{\hat{d}_\beta^4\,\tau_R}. \tag{191}$$

There are two possibilities. If $\Omega_{g*}\,\tau_{RI-SC} \ll 1$ then the plasma in the resonant layer leaves the resistive-inertial response regime when $t = \tau_{RI-SC}$, and enters the semi-collisional response regime. Moreover, the plasma leaves the resistive-inertial response regime long before the large-t state (186) is achieved. On the other hand, if $\Omega_{g*}\,\tau_{RI-SC} \gg 1$ then the plasma in the resonant layer never leaves the resistive-inertial response regime, and the large-t state (186) is eventually achieved. In the latter case, the trajectory labelled 3 in Fig. 3 terminates in the resistive-inertial response regime. In the former case, the trajectory enters the semi-collisional response regime.

In the semi-collisional response regime, Eqs. (119), (124), and (150) yield

$$\frac{\bar{\Psi}_0(g)}{E_{sb}\,\Psi_{b\,0}} = \frac{1}{\pi}\,\frac{\iota_e^{1/2}\,\hat{d}_\beta}{\tau_H\,\tau_R^{1/2}}\,\frac{1}{[g + {\rm i}\,(\omega_E + \omega_{*\,e})]\,(g + {\rm i}\,\omega_E)^{1/2}}\left(\frac{1}{g+\gamma_g} - \frac{1}{g}\right), \tag{192}$$

which can be inverse Laplace transformed to give [19]

$$\frac{\Psi_0(t)}{E_{sb}\,\Psi_{b\,0}} = \frac{1}{\pi^{3/2}}\,\frac{\iota_e^{1/2}\,\hat{d}_\beta}{\tau_H\,\tau_R^{1/2}}\int_0^t F_I(t,t')\,t'^{\,-1/2}\,{\rm e}^{-{\rm i}\,\omega_E\,t'}\,dt'. \tag{193}$$

In the small-t limit, $\gamma_g\,t$, $|\omega_{*\,e}|\,t$, $|\omega_E|\,t \ll 1$, Eq. (193) reduces to

$$\frac{\Psi_0(t)}{E_{sb}\,\Psi_{b\,0}} = \frac{\iota_e^{1/2}\,\hat{d}_\beta}{6\,\pi^{3/2}}\,\frac{\gamma_g\,t^{5/2}}{\tau_H\,\tau_R^{1/2}}. \tag{194}$$

In other words, the reconnected magnetic flux initially grows in time as $t^{5/2}$. In the large-t limit, $\gamma_g\,t$, $|\omega_{*\,e}|\,t$, $|\omega_E|\,t \gg 1$, Eq. (193) reduces to [25]

$$\frac{\Psi_0(t)}{E_{sb}\,\Psi_{b\,0}} = \frac{{\rm e}^{\gamma_g\,t}}{\Sigma_{SC}}, \tag{195}$$

where

$$\Sigma_{SC} = \frac{\pi}{\iota_e^{1/2}\,\hat{d}_\beta}\,\tau_H\,\tau_R^{1/2}\,[\gamma_g + {\rm i}\,(\omega_E + \omega_{*\,e})]\,(\gamma_g + {\rm i}\,\omega_E)^{1/2}. \tag{196}$$

Expression (195) corresponds to the contribution of the pole at $g = -\gamma_g$ when expression (192) is inverse Laplace transformed. Moreover, $\Sigma_{SC} \sim {\cal O}(S^{1/3})$ is the shielding factor.

In order for the plasma in the resonant layer to lie in the semi-collisional, rather than the diffusive-resistive, response regime, we require $\hat{\delta}_{DR} < \hat{\delta}_{SC}$. According to Eqs. (156) and (159), the plasma lies in the semi-collisional response regime provided that

$$t < \tau_{SC-DR}\,\left[1 + (\gamma_g^2 + \omega_E^2)\,t^2\right]^{1/2}, \tag{197}$$

where

$$\tau_{SC-DR} = \frac{\tau_H\,\tau_\perp^{1/2}}{\hat{d}_\beta\,\tau_R^{1/2}}. \tag{198}$$

There are two possibilities. If $\Omega_{g*}\,\tau_{SC-DR} \ll 1$ then the plasma in the resonant layer leaves the semi-collisional response regime when $t = \tau_{SC-DR}$, and enters the diffusive-resistive response regime. Moreover, the plasma leaves the semi-collisional response regime long before the large-t state (195) is achieved. On the other hand, if $\Omega_{g*}\,\tau_{SC-DR} \gg 1$ then the plasma in the resonant layer never leaves the semi-collisional response regime, and the large-t state (195) is eventually achieved. In the latter case, the trajectory labelled 3 in Fig. 3 terminates in the semi-collisional response regime. In the former case, the trajectory enters the diffusive-resistive response regime.

In the diffusive-resistive response regime, Eqs. (119), (126), and (150) yield

$$\frac{\bar{\Psi}_0(g)}{E_{sb}\,\Psi_{b\,0}} = \frac{\Gamma(1/4)}{2\pi\,\Gamma(3/4)}\,\frac{\hat{d}_\beta^{1/2}}{\tau_H^{1/2}\,\tau_R^{3/4}\,\tau_\perp^{-1/4}}\,\frac{1}{g+{\rm i}\,(\omega_E+\omega_{*\,e})}\left(\frac{1}{g+\gamma_g}-\frac{1}{g}\right), \tag{199}$$

which can be inverse Laplace transformed to give [19]

$$\frac{\Psi_0(t)}{E_{sb}\,\Psi_{b\,0}} = \frac{\Gamma(1/4)}{2\pi\,\Gamma(3/4)}\,\frac{\hat{d}_\beta^{1/2}}{\tau_H^{1/2}\,\tau_R^{3/4}\,\tau_\perp^{-1/4}}\left[\frac{{\rm e}^{\gamma_g\,t}-{\rm e}^{-{\rm i}\,(\omega_E+\omega_{*\,e})\,t}}{\gamma_g+{\rm i}\,(\omega_E+\omega_{*\,e})}-\frac{1-{\rm e}^{-{\rm i}\,(\omega_E+\omega_{*\,e})\,t}}{{\rm i}\,(\omega_E+\omega_{*\,e})}\right]. \tag{200}$$

In the small-t limit, $\gamma_g\,t$, $|\omega_{*\,e}|\,t$, $|\omega_E|\,t \ll 1$, Eq. (200) reduces to

$$\frac{\Psi_0(t)}{E_{sb}\,\Psi_{b\,0}} = \frac{\Gamma(1/4)\,\hat{d}_\beta^{1/2}}{4\pi\,\Gamma(3/4)}\,\frac{\gamma_g\,t^2}{\tau_H^{1/2}\,\tau_R^{3/4}\,\tau_\perp^{-1/4}}. \tag{201}$$

In other words, the reconnected magnetic flux initially grows in time as t^2. In the large-t limit, $\gamma_g\,t$, $|\omega_{*\,e}|\,t$, $|\omega_E|\,t \gg 1$, Eq. (200) reduces to

$$\frac{\Psi_0(t)}{E_{sb}\,\Psi_{b\,0}} = \frac{{\rm e}^{\gamma_g\,t}}{\Sigma_{DR}}, \tag{202}$$

where

$$\Sigma_{DR} = \frac{2\pi\,\Gamma(3/4)}{\Gamma(1/4)\,\hat{d}_\beta^{1/2}}\,\tau_H^{1/2}\,\tau_R^{3/4}\,\tau_\perp^{-1/4}\,[\gamma_g+{\rm i}\,(\omega_E+\omega_{*\,e})]. \tag{203}$$

Expression (203) corresponds to the contribution of the pole at $g = -\gamma_g$ when expression (199) is inverse Laplace transformed. Moreover, $\Sigma_{DR} \sim \mathcal{O}(S^{1/3})$ is the shielding factor.

G. Reconnection Scenario 4

Consider reconnection scenario 4, which corresponds to the horizontal line labelled 4 in Fig. 4. This scenario holds when $D > 1$ and $D^3 < P < D^6$.

The plasma in the resonant layer is initially in the inertial response regime. Driven magnetic reconnection in this regime is described in Sect. VI D.

If $\Omega_{g*}\,\tau_{I-VI} \ll 1$ then the plasma in the resonant layer leaves the inertial response regime when $t = \tau_{I-VI}$, and enters the viscous-inertial response regime. On the other hand, if $\Omega_{g*}\,\tau_{I-VI} \gg 1$ then the plasma in the resonant layer never leaves the inertial response regime, and the large-t state (164) is eventually achieved. In the latter case, the trajectory labelled 4 in Fig. 4 terminates in the inertial response regime. In the former case, the trajectory enters the viscous-inertial response regime.

Driven magnetic reconnection in the viscous-inertial regime is described in Sect. VI D.

In order for the plasma in the resonant layer to lie in the viscous-inertial, rather than the diffusive-inertial, response regime, we require $\hat{\delta}_{VI} < \hat{\delta}_{DI}$. According to Eqs. (154) and (155), the plasma lies in the viscous-inertial response regime provided that

$$t < \tau_{VI-DI}\,\left(1 + [\gamma_g^{\,2} + (\omega_E + \omega_{*\,e})^2]\,t^2\right)^{1/2}, \tag{204}$$

where

$$\tau_{VI-DI} = \frac{\tau_H^{\,2}\,\tau_\varphi}{\hat{d}_\beta^{\,4}\,\tau_\perp^{\,2}}. \tag{205}$$

There are two possibilities. If $\Omega_{g*}\,\tau_{VI-DI} \ll 1$ then the plasma in the resonant layer leaves the viscous-inertial response regime when $t = \tau_{VI-DI}$, and enters the diffusive-inertial response regime. Moreover, the plasma leaves the viscous-inertial response regime long before the large-t state (171) is achieved. On the other hand, if $\Omega_{g*}\,\tau_{VI-DI} \gg 1$ then the plasma in the resonant layer never leaves the viscous-inertial response regime, and the large-t state (171) is eventually achieved. In the latter case, the trajectory labelled 4 in Fig. 4 terminates in the viscous-inertial response regime. In the former case, the trajectory enters the diffusive-inertial response regime.

In the diffusive-inertial response regime, Eqs. (143), (144), and (150) yield

$$\frac{\bar{\Psi}_0(g)}{E_{sb}\,\Psi_{b\,0}} = \frac{2}{\pi}\,\frac{\hat{d}_\beta}{\tau_H\,\tau_R\,\tau_\perp^{-1/2}}\,\frac{1}{[g + {\rm i}\,(\omega_E + \omega_{*\,e})]^{3/2}}\left(\frac{1}{g + \gamma_g} - \frac{1}{g}\right), \tag{206}$$

which can be inverse Laplace transformed to give [19]

$$\frac{\Psi_0(t)}{E_{sb}\,\Psi_{b\,0}} = \frac{2}{\pi\,\Gamma(3/2)}\,\frac{\hat{d}_\beta}{\tau_H\,\tau_R\,\tau_\perp^{-1/2}}\int_0^t\left[{\rm e}^{\gamma_g\,(t-t')}-1\right]t'^{\,1/2}\,{\rm e}^{-{\rm i}\,(\omega_E+\omega_{\ast\,e})\,t'}\,dt'. \tag{207}$$

In the small-t limit, $\gamma_g\,t$, $|\omega_{\ast\,e}|\,t$, $|\omega_E|\,t\ll 1$, Eq. (207) reduces to

$$\frac{\Psi_0(t)}{E_{sb}\,\Psi_{b\,0}} = \frac{8\,\hat{d}_\beta}{15\pi\,\Gamma(3/2)}\,\frac{\gamma_g\,t^{5/2}}{\tau_H\,\tau_R\,\tau_\perp^{-1/2}}. \tag{208}$$

In other words, the reconnected magnetic flux initially grows in time as $t^{5/2}$. In the large-t limit, $\gamma_g\,t$, $|\omega_{\ast\,e}|\,t$, $|\omega_E|\,t\gg 1$, Eq. (207) reduces to [25]

$$\frac{\Psi_0(t)}{E_{sb}\,\Psi_{b\,0}} = \frac{{\rm e}^{\gamma_g\,t}}{\Sigma_{DI}}, \tag{209}$$

where

$$\Sigma_{DI} = \frac{\pi}{2\,\hat{d}_\beta}\,\tau_H\,\tau_R\,\tau_\perp^{-1/2}\left[\gamma_g+{\rm i}\,(\omega_E+\omega_{\ast\,e})\right]^{3/2}. \tag{210}$$

Expression (209) corresponds to the contribution of the pole at $g=-\gamma_g$ when expression (206) is inverse Laplace transformed. Moreover, $\Sigma_{DI}\sim{\cal O}(S^{1/3})$ is the shielding factor.

In order for the plasma in the resonant layer to lie in the diffusive-inertial, rather than the diffusive-resistive, response regime, we require $\hat{\delta}_{DI}>\hat{\delta}_{DR}$. According to Eqs. (155) and (159), the plasma lies in the diffusive-inertial response regime provided that

$$t<\tau_{DI-DR}\,\left(1+[\gamma_g^{\,2}+(\omega_E+\omega_{\ast\,e})^2]\,t^2\right)^{1/2}, \tag{211}$$

where

$$\tau_{DI-DR} = \frac{\tau_H\,\tau_R^{\,1/2}}{\hat{d}_\beta\,\tau_\perp^{\,1/2}}. \tag{212}$$

There are two possibilities. If $\Omega_{g\ast}\,\tau_{DI-DR}\ll 1$ then the plasma in the resonant layer leaves the diffusive-inertial response regime when $t=\tau_{DI-DR}$, and enters the diffusive-resistive response regime. Moreover, the plasma leaves the diffusive-inertial response regime long before the large-t state (209) is achieved. On the other hand, if $\Omega_{g\ast}\,\tau_{DI-DR}\gg 1$ then the plasma in the resonant layer never leaves the diffusive-inertial response regime, and the large-t state (209) is eventually achieved. In the latter case, the trajectory labelled 4 in Fig. 4 terminates in the diffusive-inertial response regime. In the former case, the trajectory enters the diffusive-resistive response regime.

Driven magnetic reconnection in the diffusive-resistive regime is described in Sect. VI F.

H. Reconnection Scenario 5

Consider reconnection scenario 5, which corresponds to the horizontal line labelled 5 in Fig. 4. This scenario holds when $D > 1$ and $D^2 < P < D^3$.

The plasma in the resonant layer is initially in the inertial response regime. Driven magnetic reconnection in this regime is described in Sect. VI D.

In order for the plasma in the resonant layer to lie in the inertial, rather than the diffusive-inertial, response regime, we require $\hat{\delta}_I > \hat{\delta}_{DI}$. I According to Eqs. (153) and (155), the plasma lies in the inertial response regime provided that

$$t < \tau_{I-DI}\,\frac{\left[1+(\gamma_g^2+\omega_E^2)\,t^2\right]^{1/2}\left(1+\left[\gamma_g^2+(\omega_E+\omega_{\ast\,i})^2\right]t^2\right)^{1/2}}{\left(1+\left[\gamma_g^2+(\omega_E+\omega_{\ast\,e})^2\right]t^2\right)^{1/2}}, \tag{213}$$

where

$$\tau_{I-DI} = \hat{d}_\beta^{\,2}\,\tau_\perp. \tag{214}$$

There are two possibilities. If $\varOmega_{g\ast}\,\tau_{I-DI} \ll 1$ then the plasma in the resonant layer leaves the inertial response regime when $t = \tau_{I-DI}$, and enters the diffusive-inertial response regime. Moreover, the plasma leaves the inertial response regime long before the large-t state (164) is achieved. On the other hand, if $\varOmega\,\tau_{I-DI} \gg 1$ then the plasma in the resonant layer never leaves the inertial response regime, and the large-t state (164) is eventually achieved. In the latter case, the trajectory labelled 5 in Fig. 4 terminates in the inertial response regime. In the former case, the trajectory enter the diffusive-inertial response regime.

Driven magnetic reconnection in the diffusive-inertial regime is described in Sect. VI G.

If $\varOmega_{g\ast}\,\tau_{DI-DR} \ll 1$ then the plasma in the resonant layer leaves the diffusive-inertial response regime when $t = \tau_{DI-DR}$, and enters the diffusive-resistive response regime. On the other hand, if $\varOmega_{g\ast}\,\tau_{DI-DR} \gg 1$ then the plasma in the resonant layer never leaves the diffusive-inertial response regime, and the large-t state (209) is eventually achieved. In the latter case, the trajectory labelled 5 in Fig. 4 terminates in the diffusive-inertial response regime. In the former case, the trajectory enters the diffusive-resistive response regime.

Driven magnetic reconnection in the diffusive-resistive regime is described in Sect. VI F.

I. Reconnection Scenario 6

Consider reconnection scenario 6, which corresponds to the horizontal line labelled 6 in Fig. 4. This scenario holds when $D > 1$ and $D^{-6} < P < D^2$.

The plasma in the resonant layer is initially in the inertial response regime. Driven magnetic reconnection in this response regime is described in Sect. VI D.

In order for the plasma in the resonant layer to lie in the inertial, rather than the diffusive-resistive, response regime, we require $\hat{\delta}_I > \hat{\delta}_{DR}$. I According to Eqs. (153) and (159), the plasma lies in the inertial response regime provided that

$$t < \tau_{I-DR}\,\left[1+(\gamma_g^{\,2}+\omega_E^{\,2})\,t^2\right]^{1/4}\left(1+\left[\gamma_g^{\,2}+(\omega_E+\omega_{\ast\,i})^2\right]t^2\right)^{1/4} \tag{215}$$

where

$$\tau_{I-DR} = \hat{d}_\beta^{\,1/2}\,\tau_H^{\,1/2}\,\tau_R^{\,1/4}\,\tau_\perp^{\,1/4}. \tag{216}$$

There are two possibilities. If $\varOmega_{g\ast}\,\tau_{I-DR} \ll 1$ then the plasma in the resonant layer leaves the inertial response regime when $t = \tau_{I-DR}$, and enters the diffusive-resistive response regime. Moreover, the plasma leaves the inertial regime long before the large-t state (164) is achieved. On the other hand, if $\varOmega_{g\ast}\,\tau_{I-DR} \gg 1$ then the plasma in the resonant layer never leaves the inertial response regime, and the large-t state (164) is eventually achieved. In the latter case, the trajectory labelled 6 in Fig. 4 terminates in the inertial response regime. In the former case, the trajectory enter the diffusive-resistive response regime.

Driven magnetic reconnection in the diffusive-resistive response regime is described in Sect. VI F.

J. Reconnection Scenario 7

Consider reconnection scenario 7, which corresponds to the horizontal line labelled 7 in Fig. 4. This scenario holds when $D > 1$ and $P < D^{-6}$.

The plasma in the resonant layer is initially in the inertial response regime. Driven magnetic reconnection in this response regime is described in Sect. VI D.

In order for the plasma in the resonant layer to lie in the inertial, rather than the semi-collisional, response regime, we require $\hat{\delta}_I > \hat{\delta}_{SC}$. According to Eqs. (153) and (156), the plasma lies in the inertial response regime provided that

$$t < \tau_{I-SC}\,\left(1 + [\gamma_g^{\,2} + (\omega_E + \omega_{\ast\,i})^2]\,t^2\right)^{1/2}, \tag{217}$$

where

$$\tau_{I-SC} = \hat{d}_\beta^{\,2}\,\tau_R. \tag{218}$$

There are two possibilities. If $\varOmega_{g\ast}\,\tau_{I-SC} \ll 1$ then the plasma in the resonant layer leaves the inertial response regime when $t = \tau_{I-SC}$, and enters the semi-collsional response regime. On the other hand, if $\varOmega_{g\ast}\,\tau_{I-SC} \gg 1$ then the plasma in the resonant layer never leaves the inertial response regime, and the large-t state (164) is eventually achieved. In the latter case, the trajectory labelled 7 in Fig. 4 terminates in the inertial response regime. In the former case, the trajectory enter the semi-collisional response regime.

Driven magnetic reconnection in the semi-collisional response regime is described in Sect. VI F.

If $\varOmega_{g\ast}\,\tau_{SC-DR} \ll 1$ then the plasma in the resonant layer leaves the semi-collisional response regime when $t = \tau_{SC-DR}$, and enters the diffusive-resistive response regime. On the other hand, if $\varOmega_{g\ast}\,\tau_{SC-DR} \gg 1$ then the plasma in the resonant layer never leaves the semi-collisional response regime, and the large-t state (195) is eventually achieved. In the latter case, the trajectory labelled 7 in Fig. 4 terminates in the semi-collisional response regime. In the former case, the trajectory enters the diffusive-resistive response regime.

Driven magnetic reconnection in the diffusive-resistive response regime is described in Sect. VI F.

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DATA AVAILABILITY STATEMENT

The digital data used in the figures in this paper can be obtained from the author upon reasonable request.

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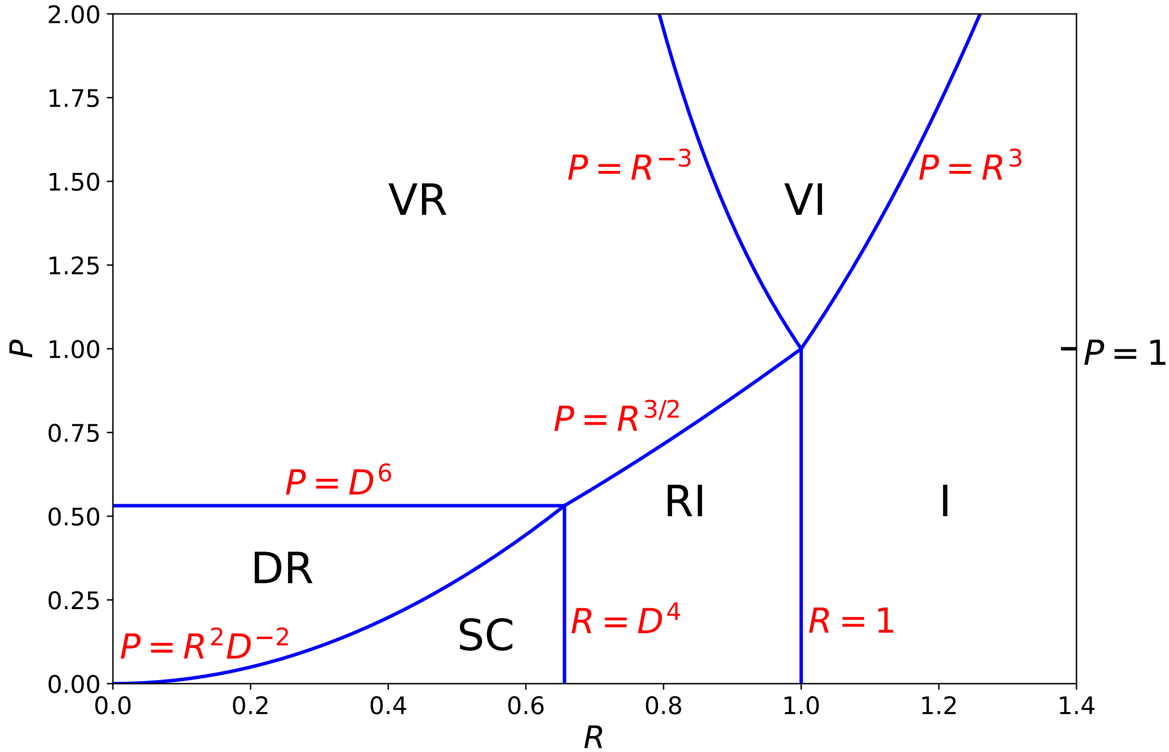

FIG. 1. Linear resonant plasma response regimes in R-P space for the case $D = 0.9$. The various regimes are the diffusive-resisitive (DR), the semi-collisional (SC), the resistive-inertial (RI), the viscous-resistive (VR), the viscous-inertial (VI), and the inertial (I).

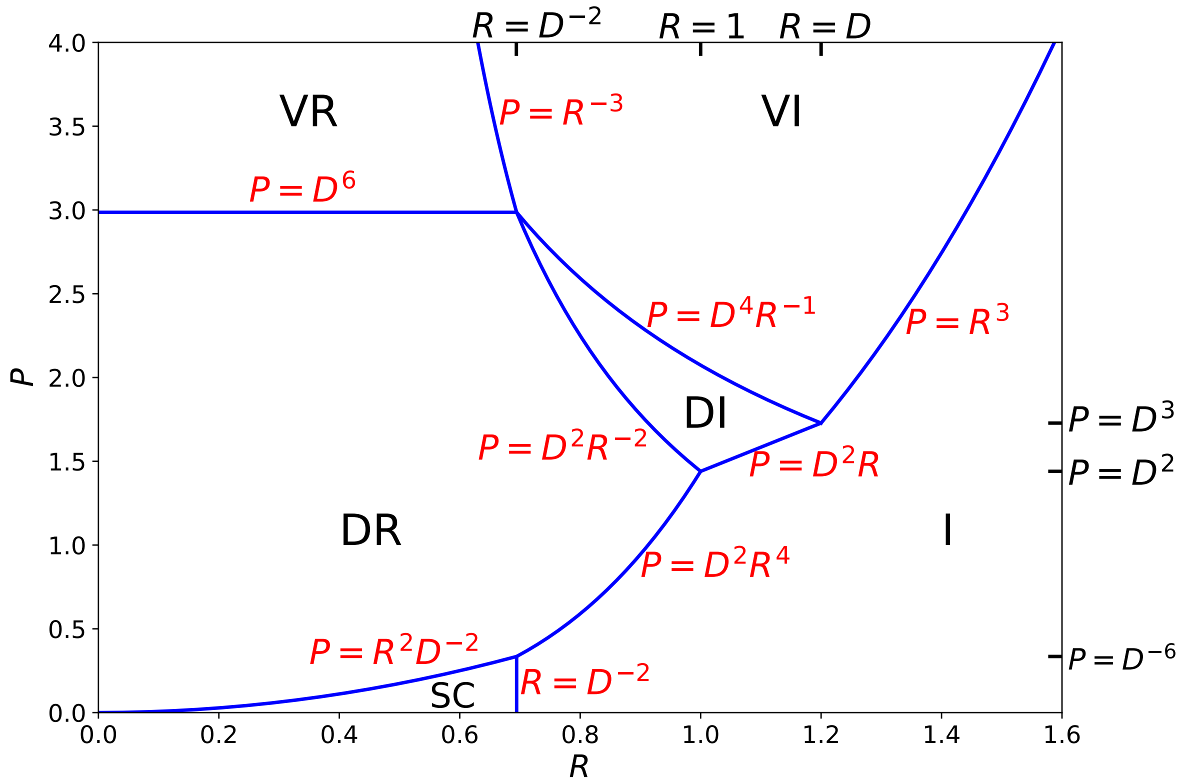

FIG. 2. Linear resonant plasma response regimes in R-P space for the case $D = 1.2$. The various regimes are the diffusive-resisitive (DR), the semi-collisional (SC), the diffusive-inertial (DI), the viscous-resistive (VR), the viscous-inertial (VI), and the inertial (I).

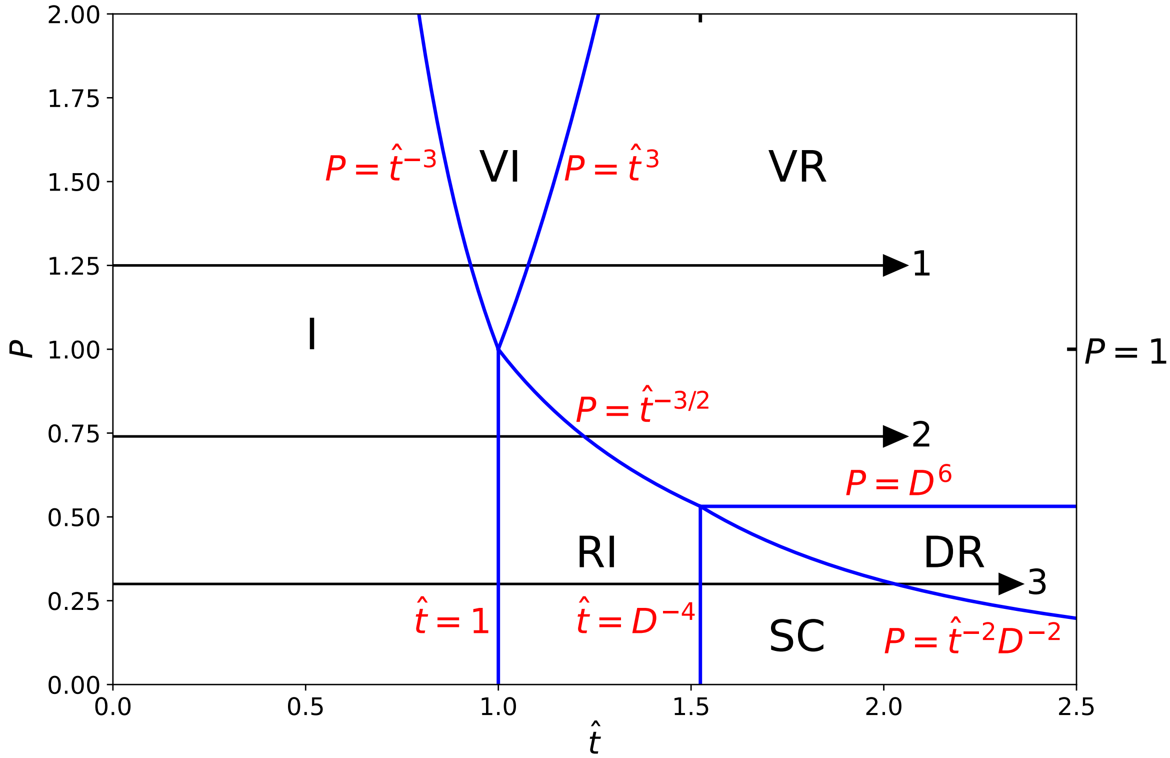

FIG. 3. Trajectory of the Fourier/Laplace transformed response of a resonant layer to a resonant magnetic perturbation in $\hat{t}$-P space for the case $D = 0.9$. The various regimes are the diffusive-resistive (DR), the semi-collisional (SC), the resistive-inertial (RI), the viscous-resistive (VR), the viscous-inertial (VI), and the inertial (I).

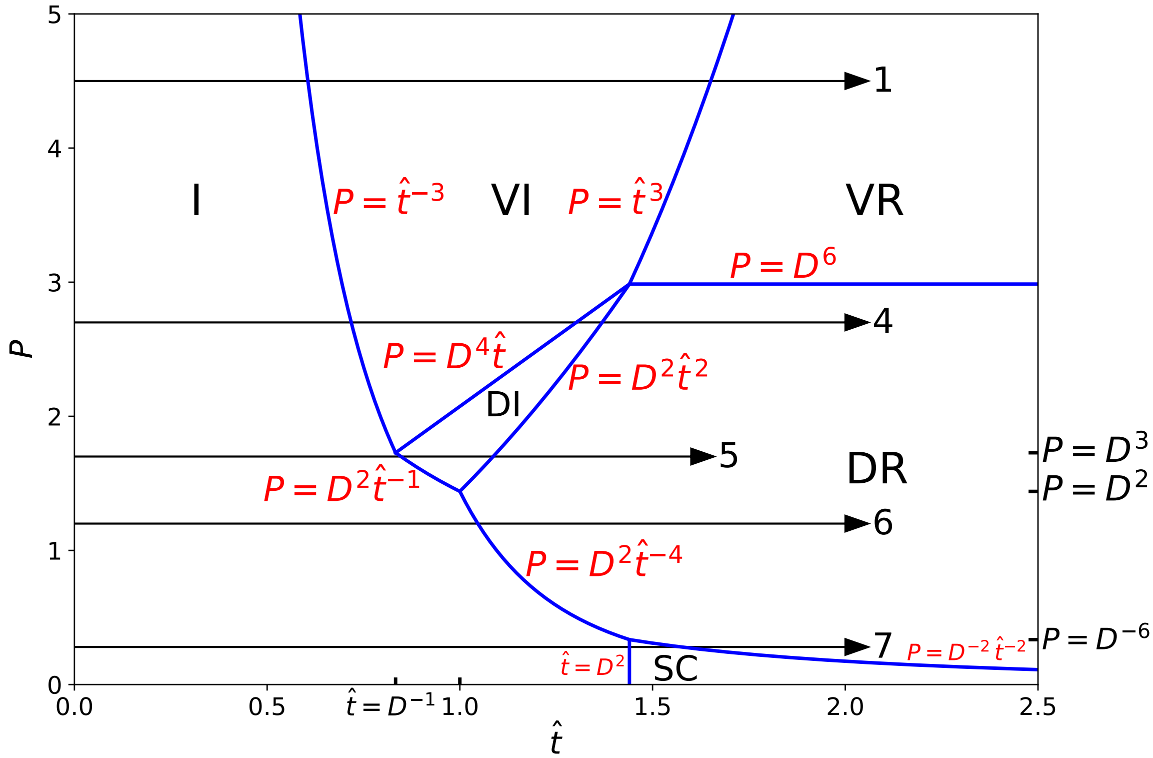

FIG. 4. Trajectory of the Fourier/Laplace transformed response of a resonant layer to a resonant magnetic perturbation in $\hat{t}$-P space for the case $D = 1.2$. The various regimes are the diffusive-resistive (DR), the semi-collisional (SC), the diffusive-inertial (DI), the viscous-resistive (VR), the viscous-inertial (VI), and the inertial (I).